Journal of Cardiac Failure 00 (2026) 1–23

JCF
Journal of Cardiac Failure



HFSA Scientific Statement

Heart Failure Society of America Scientific Statement on Right Heart Failure

MANREET K. KANWAR, MD[1] ANTHONY P. CARNICELLI, MD[2] MARK N. BELKIN, MD[1] MICHELE L. ESPOSITO, MD[2] JAMES C. COONS, PharmD[3] EZEQUIEL J. MOLINA, MD[4] SOPHIA AIRHART, MD[5] THOMAS M. CASCINO, MD, MSc[6] VANESSA BLUMER, MD[7] and RYAN J. TEDFORD, MD[2]

HFSA Scientific Statement Committee Members

David Lanfear, MD, *Chair*

Johanna Contreras, MD
Corrine Jurgens, PhD, RN
Kazuhiko Kido, PharmD
Shannon Dunlay, MD
Senthil Selvaraj, MD
Justin Vader, MD

ABSTRACT

Right heart failure (RHF), both acute and chronic, is increasingly recognized as a distinct and clinically significant syndrome across the spectrum of cardiovascular disease. Estimates suggest RHF complicates >25%–30% of hospitalizations for acute decompensated heart failure and is nearly universal in advanced pulmonary hypertension. Despite recent diagnostic and therapeutic advancements, RHF remains associated with substantial morbidity, mortality, and impaired quality of life. Addressing RHF requires dedicated strategies tailored to its pathophysiology, including early identification, risk stratification, and novel therapies. Although randomized controlled trials on the topic remain limited, there has been increasing emphasis on early recognition and management of RHF (especially in the setting of left ventricular failure). In addition, there is an expanding role for temporary mechanical circulatory support in acute RHF and percutaneous tricuspid interventions in chronic RHF. Yet, the field remains limited by its underrecognition, lack of standardized definitions, and limited therapeutic options, which underscore the urgent need for consensus and guidance (*J Cardiac Fail 2026;00:1–23*)

Key Words: Right heart failure, pulmonary hypertension, cardiogenic shock.

© 2026 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

Introduction

Right heart failure (RHF), both acute and chronic, is increasingly recognized as a distinct and clinically significant syndrome across the spectrum of cardiovascular disease. It complicates >25%–30% of hospitalizations for acute decompensated HF and is nearly universal in advanced pulmonary hypertension (PH).[1] Despite recent

From the [1]Division of Cardiology, Department of Medicine, University of Chicago, Chicago, Illinois; [2]Division of Cardiology, Department of Medicine, Medical University of South Carolina, Charleston, South Carolina; [3]UPMC Presbyterian Hospital, University of Pittsburgh School of Pharmacy, Pittsburgh, Pennsylvania; [4]Piedmont Heart Institute, Atlanta, Georgia; [5]Division of Cardiology, Department of Medicine, Baystate Medical Center, University of Massachusetts Chan Medical School-Baystate, Springfield, Massachusetts; [6]Division of Cardiology, University of Michigan, Ann Arbor, Michigan and [7]Inova Schar Heart and Vascular, Falls Church, Virginia.

Manuscript received October 14, 2025; revised manuscript received April 18, 2026; revised manuscript accepted April 27, 2026.

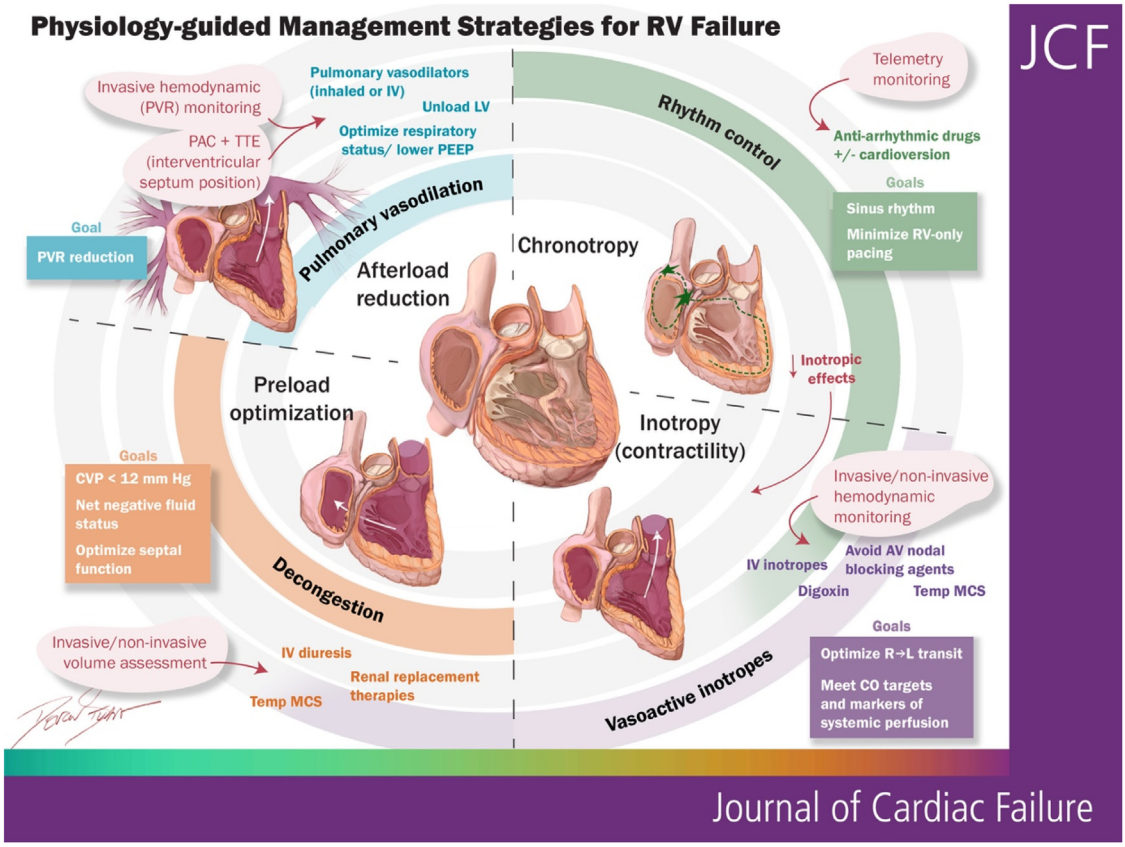
Reprint requests: Manreet Kanwar, MD, 5841 South Maryland Ave, Chicago, IL 60637. E-mail: manreetkanwar@uchicago.edu

See page 19 for disclosure information.

1071-9164/$ - see front matter

© 2026 Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

https://doi.org/10.1016/j.cardfail.2026.04.026



Central Illustration. Physiology-guided management strategies for right ventricular (RV) failure. AV, aortoventricular; CO, cardiac output; CVP, central venous pressure; IV, intravenous; LV, left ventricle; MCS, mechanical circulatory support; PAC, pulmonary artery catheter; PEEP, positive end expiratory pressure; PVR, pulmonary vascular resistance; RV, right ventricle; TTE, transthoracic echocardiogram.

diagnostic and therapeutic advancements, RHF remains associated with substantial morbidity, mortality, and impaired quality of life.

Addressing RHF requires dedicated strategies tailored to its pathophysiology, including early identification, risk stratification, and novel therapies.[2] Although randomized controlled trials on the topic remain limited, there has been increasing emphasis on early recognition and management of RHF (especially in the setting of left ventricular [LV] failure). In addition, there is an expanding role for temporary mechanical circulatory support (tMCS) in acute RHF and percutaneous tricuspid interventions in chronic RHF.

The Heart Failure Society of America convened this Scientific Statement to synthesize current evidence, identify knowledge gaps, and promote best practices in the recognition and management of RHF. The last Scientific Statement on evaluation and management of RHF was published by the American Heart Association in 2018 and focused on the epidemiology, mechanisms, diagnostic tools, and therapeutic strategies.[1] Our writing group focused on systematic literature reviews, translational and clinical studies, clinical practice guidelines, and expert statements between 2018 and 2025 with the aim of summarizing current evidence and highlighting gaps that warrant future research in RHF. Specifically, we aim to focus on the approach to RHF management in different clinical states and not on underlying disease state management. By positioning RHF as a distinct, prognostically meaningful syndrome, this statement calls for a paradigm shift: from viewing RHF as a secondary complication to recognizing it as a central driver of outcomes in diverse patient populations.

Defining RHF

RHF is defined as a clinical syndrome resulting from dysfunction within the right heart circulatory system. The right ventricle (RV) represents one major component of this system; however, other components of the right heart system, such as the right atrium (RA) and pulmonary arterial vasculature, play important roles in some etiologies of RHF. Here, we primarily examine the pathophysiology of the RV as it contributes to RHF.

The RV's principal functions are to generate forward flow into the pulmonary circulation, offload systemic venous

return, and provide sufficient LV preload. Failure in 1 or more of these functions can lead to elevated systemic venous pressures, end-organ hypoperfusion, inadequate cardiac output (CO), and hemodynamic compromise, manifesting clinically as signs and symptoms of RHF.

Mechanisms and Pathophysiology

The pathophysiology of RV failure encompasses several mechanisms, which can be categorized as follows:

(1) Excessive preload: leads to RV dilatation, increased wall tension, altered septal geometry, increased ventricular interdependence, decreased coronary perfusion pressure, myocardial ischemia, tricuspid regurgitation (TR), and reduced contractile reserve and efficiency.
(2) Excessive afterload: leads to altered RV geometry, increased wall tension, RV-pulmonary arterial uncoupling, RV remodeling, TR, and excessive preload.
(3) Reduced inotropy: leads to impaired RV systolic function, impaired LV filling, RV remodeling, ventricular desynchrony, RV-pulmonary arterial uncoupling, and excessive preload.
(4) Reduced lusitropy: leads to increased ventricular interdependence, altered RV geometry, and excessive preload.

Most cases of RV failure begin with one primary mechanism, but over time other mechanisms typically evolve and compound the progression of the disease. Specific etiologies of RV failure with their associated primary mechanisms are listed in Table 1.

Table 1 Specific etiologies of right ventricular failure based on underlying primary mechanism

Mechanism	Etiology
Excessive preload	Tricuspid regurgitation Pulmonic regurgitation Congestion from left heart failure Left-to-right intracardiac shunts Extracardiac arteriovenous shunts
Excessive afterload	Mechanical obstruction (pulmonary embolism, pulmonic stenosis, congenital heart disease) Precapillary pulmonary hypertension Lung disease and hypoxia Postcapillary pulmonary hypertension
Reduced inotropy	Myocardial infarction Myocarditis Post-LVAD or cardiac surgery Arrhythmogenic right ventricular cardiomyopathy Infiltrative cardiomyopathies Systemic sclerosis
Reduced lusitropy	Myocardial fibrosis Sarcomere stiffening Infiltrative cardiomyopathies Constrictive pericarditis

LVAD, left ventricular assist device.

Epidemiology

RHF epidemiology varies according to the clinical scenario and underlying etiology:

Cardiogenic shock. RV-predominant cardiogenic shock represents only 5% of patients with acute myocardial infarction–related cardiogenic shock (AMI-CS). However, concomitant LV and RHF, leading to biventricular shock, is significantly more common.[3,4] Contemporary data consistently show that RV dysfunction is prevalent in 30%–40% of patients with AMI-CS and a similar percentage of those with acute-on-chronic HF CS. Both clinical RHF and RV dysfunction in AMI-CS portend worse survival.

Pulmonary arterial hypertension (PAH). RHF is highly prevalent in PAH, present in up to 50% of patients at diagnosis, and nearly universal in advanced disease. Registry data from the Registry to Evaluate Early and Long-Term Pulmonary Arterial Hypertension Disease Management (REVEAL), the largest US-based PAH registry, demonstrate that RHF is responsible for 56% of hospitalizations in PAH and that RHF is the leading proximate cause of death in the majority of cases. The presence of RV dysfunction has been shown in both REVEAL and Comparative, Prospective Registry of Newly Initiated Therapies for Pulmonary Hypertension (COMPERA) registries to significantly increase the predictive ability of previously established risk scores.

Chronic left heart failure (HF). RV dysfunction commonly complicates chronic left-sided HF. Observational data suggest that RV dysfunction is present in approximately 50% of patients with chronic HF, regardless of whether LV ejection fraction is preserved or reduced. Severe decompensated LV failure will often be accompanied by many of the clinical signs and symptoms of RHF due to excessive preload described previously. This overlap complicates the distinction between isolated/contributory RHF and secondary RV dysfunction due to LV failure, thus presenting challenges in quantifying RHF epidemiology in this population.

Perioperative RHF. RHF develops in approximately 20% of patients after durable LVAD implantation and may develop early (<30 days) or late (>30 days). Primary isolated early RHF after orthotopic heart transplantation occurs in up to 20% of cases and is associated with poor outcomes.

Clinical manifestations and diagnosis of RHF

Signs, Symptoms, and Diagnostic Work-up

Advanced HF specialists are well acquainted with the clinical findings of RHF. Patients may present with hypoperfusion and hypotension, particularly when acute, which in

turn results in a cascade of organ injury via reduction in tissue perfusion pressure. Most commonly recognized is the effect on renal function, because congestive nephropathy often leads to acute kidney injury, progressive diuretic resistance, and in some cases oliguria/anuria necessitating renal-replacement therapy.[5] The liver is similarly affected, as venous hypertension leads to congestive hepatopathy and intrahepatic cholestasis, ultimately progressing to coagulopathy and altered drug metabolism in severe cases.[6] Acute splanchnic congestion in this setting may result in gut-wall edema, resulting in slowed gastrointestinal transit and cachexia. When compounded by hypoperfusion, this can lead to ischemic bowel and impaired mucosal barrier function, which facilitates bacterial translocation, increases the risk of sepsis/bacteremia, and increases exposure to bacterial endotoxin, thus amplifying the systemic inflammatory response.[6] In chronic RHF, symptoms are often less specific and may be misattributed to, or confounded by, noncardiac comorbidities or left HF, delaying diagnosis and treatment. Hematologic abnormalities also may develop secondary to RHF, most commonly anemia secondary to chronic inflammation, impaired erythropoietin production from renal dysfunction, and abnormal iron uptake/handling.[7]

When RHF is suspected, history should focus on identification of etiologies and characterization of disease duration and markers of severity. Diagnostic evaluation should integrate clinical examination, laboratory evaluation, imaging, and hemodynamics.[8-19] Initial laboratory evaluation includes assessment of markers of noncardiac organ function (predominantly renal and hepatic) as well as markers of disease severity and prognosis (Table 2).[8-15,17-19,20-23] This evaluation should be guided by a physical examination and imaging.

Transthoracic echocardiography is the first-line imaging modality and an indispensable tool to assess the right atrium (RA) and RV size and function, ventricular interdependence, tricuspid valve competence, and pulmonary pressures. The American Society of Echocardiography released updated guidelines on the assessment of the right heart with a focus on structural and functional right heart adaptation to increased afterload as well as the essential role for transthoracic echocardiography in screening, detection, and prognostication in PH.[20] Key echocardiographic and hemodynamic measures of RV dysfunction and RV-pulmonary artery (PA) coupling are summarized in Table 3.[21-45] Maladaptive RV remodeling is characterized by echocardiographic evidence of RA and RV dilatation, worsening RV systolic function, and RV-PA uncoupling, parameters that have demonstrated consistent prognostic value across disease states. RV systolic function can be assessed by RV fractional area change, tricuspid annular plane systolic excursion (TAPSE), tissue Doppler S′ velocity, and 3-dimensional RV ejection fraction. RV fractional area change reflects global RV systolic function and correlates well with cardiac magnetic resonance–derived RVEF, although it does not capture RV outflow tract contribution and is dependent on image quality and endocardial definition. Longitudinal measures such as TAPSE and S′ are prognostic but are angle- and load-dependent and primarily reflect longitudinal contraction, underscoring the importance of integrating multiple complementary indices of RV function rather than relying on any single parameter. RV-PA coupling metrics that integrate RV contractile performance with afterload, particularly the TAPSE/PA systolic pressure ratio and RV strain, provide robust risk stratification and identify maladaptive uncoupling. Additional high-risk echocardiographic features include worsening tricuspid regurgitation, abnormal eccentricity index, pericardial effusion, and disproportionate right-sided remodeling and estimated filling pressures relative to the left ventricle.

Interpretation of RV indices requires clinical context, because normal reference values for many RV measures remain incompletely standardized and vary by acquisition technique, vendor and software, study population, and patient factors such as sex, age, and body size. Previous cardiac surgery may further confound assessment by altering annular motion and RV geometry, limiting the reliability of TAPSE and related measures independent of intrinsic RV systolic function. In such settings, cardiac magnetic resonance imaging may provide superior assessment of RV volumes, geometry, and systolic function and offer additional tissue characterization, including detection of fibrosis and infiltration, making it particularly valuable in congenital heart disease and infiltrative cardiomyopathies or when echo assessment is limited or discordant with clinical findings.[30,46]

Right heart catheterization remains essential for hemodynamic profiling, PH phenotyping, and risk stratification to guide therapy.[26,30,33,38,47-55] The acute and chronic changes exhibited by the dysfunctional RV can be captured from the data obtained from a PA catheter, which might be helpful in guiding decisions on clinical management or prognostication. Initially, acute changes are typically signaled by elevated right-sided filling pressures, particularly right atrial pressure (RAP) or the RA/pulmonary artery wedge pressure (PAWP) ratio (Table 4). Although an elevated RAP does not in and of itself differentiate between right- and left-sided heart failure, a high RA/PAWP ratio might signal more predominant RHF physiology, as compared with a biventricular failure phenotype.[56] PAWP is also a central marker for distinguishing pre- from postcapillary PH. Although both forms of PH may lead to RHF, therapeutic strategies are markedly different. However, technical limitations of PAWP measurement—such as respiratory variation, transducer leveling, and incomplete occlusion—combined with physiologic variability (eg, obesity, V-waves, impaired compliance) may undermine its accuracy. Rayner et al[57] proposed a "zone of uncertainty" between 12 and 18 mm Hg, where PAWP alone lacks the precision to differentiate predominant left-

Table 2 Clinical and biomarker predictors of outcomes in RHF

Parameter	Cut-offs/Thresholds	Risk Interpretation	Supporting Data/Notes	Limitations
Clinical				
WHO/NYHA functional class[8,9]	I → IV	Low vs high risk	Consistently associated with mortality and hospitalization in PAH registries.	Subjective; interobserver variability.
6-min walk distance[20-23]	>440 m (low) | <165 m (high)	Low value correlates with mortality and hospitalization in a continuous fashion	Validated as a prognostic marker in idiopathic PAH Confirmed in large registries. Improvement ≥30−40 m is clinically meaningful.	Influenced by musculoskeletal disease, age, sex, effort, and comorbidities; limited reproducibility.
Biomarkers/laboratory data				
BNP NT-proBNP[20-23]	<50/>800 ng/mL <300/>1100 ng/L	Low vs high risk	Reflects RV wall stress	Renal dysfunction, atrial fibrillation can confound.
Troponin (I or T, high-sensitivity)[10,11]	Positive, adverse	Greater risk	In PAH, detectable cTnI a/w ~4.7-fold greater risk at 3 years; detectable cTnT independently predicted death in chronic pre-capillary PH.	Assay-specific; overlap with left heart disease or ischemia, non-specific myocardial injury.
Liver function (Tbili), albumin[12,13]	Tbili >1.2 mg/dL adverse; low albumin adverse	Greater risk	Hepatic congestion, low synthetic function correlate with outcomes.	Nonspecific; liver disease confounds.
Renal function[14]	eGFR<60 mL/min/1.73 m^2, Cr >1.4 mg/dL adverse	Greater risk	Cardiorenal association with decompensation	Nonspecific; renal disease confounds.
Sodium[17]	<136 mmol/L adverse	Greater risk	Hyponatremia tracks neurohormonal activation and prognosis.	Volume status and diuretics confound.
Uric acid[15,18]	Continuous risk with increased levels. Pragmatic thresholds: >7 mg/dL (men) or >6 mg/dL (women) a/w higher mortality and advanced hemodynamic derangement	Greater uric acid correlates with greater RAP, and increased risk of death; each 1 mg/dL increase confers ~15%−20% greater mortality risk.	Validated in idiopathic and connective-tissue PAH cohorts; reflects oxidative stress and impaired purine metabolism under low CO.	Nonspecific; diet/renal confound.
Arterial P_{CO_2}[19]	Continuous risk; ≤34 mm Hg associated with adverse prognosis	Lower P_{aCO_2} reflects greater ventilatory inefficiency, greater PVR, and lower CO/low perfusion	Independent predictor in multiple cohorts; correlates with VE/V_{CO_2} slope and NYHA class.	Respiratory comorbidities confound. Can be influenced by anxiety, sampling technique, or hyperventilation unrelated to PH severity.

Please note that the stated thresholds are derived from reference clinical studies identifying adverse outcomes in persons with RHF.

BNP, brain natriuretic peptide; CO, cardiac output; Cr, creatinine; cTnI, cardiac troponin I; cTnT, cardiac troponin T; eGFR, estimated glomerular filtration rate; NT-proBNP, N-terminal pro B-type natriuretic peptide; NYHA, New York Heart Association; P_{aCO_2}, partial pressure of carbon dioxide; PAH, pulmonary arterial hypertension; P_{CO_2}, partial pressure of carbon dioxide; PH, pulmonary hypertension; PVR, pulmonary vascular resistance; RAP, right atrial pressure; RHF, right heart failure; Tbili, total bilirubin; Tn, troponin; VE/V_{CO_2}, ventilation/carbon dioxide production; WHO, World Health Organization.

ARTICLE IN PRESS

Table 3 Imaging parameters that predict outcomes in RHF

Parameter	Cut-offs/Thresholds	Risk Interpretation	Supporting Data/Notes	Limitations
Echocardiographic parameters				
RV/LV basal diameter ratio[30]	>1.0 abnormal	Greater risk	RV remodeling associated with mortality.	Plane/foreshortening sensitivity
RVESRI (end-systolic remodeling index)[37]	≥1.6 adverse	Greater risk	Predicts outcomes in PAH.	
RA area (end-systole)[40]	>18 cm^2 adverse	Greater risk	Chronic RV overload marker.	Load dependent
ΔRA area,[34] cm^2	−1.30	Decreased risk	Tracks remodeling	
ΔRVEDA,[34] cm^2	−2.45	Decreased risk	Tracks remodeling	
Severe TR[35]		Increased risk		
TAPSE[31,36,39]	Normal range 24 ± 3.5 mm PAH <18 mm was associated with a poor prognosis; for every 1-mm decrease in TAPSE, the unadjusted risk of death increased by 17%. <14 mm associated with all-cause death and HF hospitalization in HFrEF and HFpEF.	Greater risk	Independent predictor of mortality.	Angle/load dependent. Reflects primarily longitudinal function. Should be combined with other measures of RV function that assess radial/anteroposterior shortening.
TAPSE/PASP (RV-PA coupling)[22-25]	Continuous <0.19 mm/mm Hg high risk in PAH; >0.32 mm/mm Hg predicted low 1-y mortality in PH, TAPSE/sPAP ≤0.55 mm/mm Hg predicts poor outcomes with TAVR, ≤0.41 mm/mm Hg predicts poor outcomes with transcatheter TVI		Cut off varies by population.	Not feasible if no TR jet.
RVFWS/PASP[30]	<0.19%/mm Hg predicted adverse outcomes			
RV FAC[34]	Normal range 49% ± 7% <35%−36.5% adverse	Greater risk	Fair correlation with CMR-derived RV EF.	Image quality dependent. Poor reproducibility. Neglects contribution of the RVOT. Load dependent.
3D RVEF[28]	In PAH <38% had significantly shorter event-free survival. In DCM, <43.4% predicted composite endpoint of cardiac death, nonfatal cardiac arrest, and acute worsening of HF requiring hospitalization intermediate- to high-risk stratification.			

(*Continued*)

Table 3 (*Continued*)

Parameter	Cut-offs/Thresholds	Risk Interpretation	Supporting Data/Notes	Limitations
S′ (TDI)	Normal range 14.1 ± 2.3 cm/s Abnormal threshold <9.5 cm/s associated with increased 1-y all-cause mortality in patients undergoing TAVR.	Greater risk	Correlates with RVEF and outcomes.	Reflects only longitudinal function of basal lateral wall Load and angle dependent. should be combined with other measures of RV function that assess circumferential/antero-posterior shortening.
Pericardial effusion[54]	Present, adverse	Greater risk	Associated with advanced RV failure and mortality.	Nonspecific etiologies.
RV dyssynchrony (RV-SD4)[34]	>23 ms adverse; ≤18 ms low risk		Associated with outcomes; reversible with therapy.	Strain software variability.
Cardiac MRI[30,46]				
RVEDVi	>84 mL/m^2 adverse in iPAH	Greater risk	Morphologic remodeling associated with mortality.	Indexing and vendor differences.
LVEDVi	≤40 increased risk; >58 decreased risk in PAH			
RVESVi, % pred	RVESVI ≥180% of predicted independently predicted mortality at 1 and 3 y in PAH. RVESVI >227% predicted was the optimal threshold for severe adverse remodeling and worst survival quartile	Greater risk	Thresholds stratify risk in PAH.	Predicted values vary by package.
RV M/V ratio g/mL	≤0.45 increased risk in PAH			
RVEF	<37% predicts high (>10%) 1-y mortality in PAH. ≤45% associated with composite (all-cause mortality or cardiac transplantation) in DCM	Greater risk	Among strongest CMR predictors of mortality.	Depends on segmentation quality.
ΔRVEF over time	Any decline, adverse	Worsening risk	Serial improvement associates with better outcomes.	Requires repeat CMR.
SVI	≤25 mL/m^2 adverse	Greater risk	Lower SVI associated with mortality.	Highly preload/afterload sensitive.
RV SV/ESV	≤0.53 adverse	Greater risk (uncoupled)	Coupling metric predicting outcomes.	Less validated outside PAH.

Please note that the stated thresholds are derived from reference clinical studies identifying adverse outcomes in persons with RHF.

CMR, cardiac magnetic resonance; DCM, dilated cardiomyopathy; FAC, fractional area change; HF, heart failure; HFpEF, heart failure with preserved ejection fraction; HFrEF, heart failure with reduced ejection fraction; LV, left ventricular; LVEDVi, left ventricular end-diastolic volume indexed to body surface area; M/V, mass-to-volume; PAH, pulmonary arterial hypertension; PASP, pulmonary artery systolic pressure; PH, pulmonary hypertension; RA, right atrial; RHF, right heart failure; RV, right ventricular; RVEDA, right ventricular end-diastolic area; RVEDVi, right ventricular end-diastolic volume indexed to body surface area; RVEF, right ventricular ejection fraction; RVESRI, right ventricular end-systolic remodeling index; RVESVi, right ventricular end-systolic volume indexed to body surface area; RVFWS, right ventricular free-wall strain; RVOT, right ventricular outflow tract; RV-SD4, right ventricular 4-segment standard deviation of time to peak strain; SV/ESV, stroke volume/end-systolic volume; S′, peak systolic tricuspid annular velocity; SVI, stroke-volume index; TAPSE, tricuspid annular plane systolic excursion; TAVR, transcatheter aortic valve replacement; TDI, tissue Doppler imaging; TR, tricuspid regurgitation; TVI, tricuspid valve intervention.

ARTICLE IN PRESS

Table 4 Hemodynamic predictors of outcomes in RHF

Parameter	Cut-offs/Thresholds	Risk Interpretation	Supporting Data/Notes	Limitations
Hemodynamics				
Right atrial pressure (RAP)[26,30]	<8 (low) | >14 mm Hg (high)	Low vs high risk	Strong independent prognostic marker.	Volume dependent.
RA/PCWP ratio[56]	≥0.63 (adverse); <0.4 (adaptive)	Greater ratio indicates predominant RHF physiology and venous congestion	Validated in AHF and PH cohorts; correlates with RV dilatation, low CI, and poor outcomes. Useful in differentiating isolated RHF vs biventricular failure.	Load-dependent; PCWP accuracy affects ratio.
RVSWi (mean PAP − RAP) × SVI × 0.0136[51-53]	<5−7 g·m/m^2/beat (adverse)	Lower RVSWI reflects impaired RV contractile performance	Validated in PAH and LVAD populations	Requires accurate TD SVI; low values predict mortality and postoperative RHF.
PAPi (PASP-PADP)/RAP[60-62]	<1.5, high risk; 1.5−3, intermediate; >3, low risk	Low PAPi indicates RV-PA uncoupling and elevated RHF risk	Validated in cardiogenic shock, LVAD, and PH.	Population and preload dependent.
RV-MPS[182]	<0.5, high risk, ≥0.5, intermediate-low risk	Low RV-MPS indicated RV-PA uncoupling, reduced contractility and efficiency,	Validated in silico and in AHF population; associated with postoperative RVF after LVAD.	More data needed for this novel metric.
PVR[54]	Continuous variable; PVR >5 WU often used as high-risk cutoff	Greater PVR reflects more severe pulmonary vascular remodeling and higher RV afterload,	Shown in PH registries that PVR >5 WU stratifies worse survival. Also prognostic value of residual PVR posttherapy,	Flow-dependent (inversely influenced by CO), affected by vasoreactivity and measurement error; "intermediate" PVR cutoffs are less well validated.
CI[30,55]	≥2.5 (low) | <2.0 L/min • m^2 (high)	Low vs high risk	Survival discriminator across registries.	Thermodilution/Fick method differences.
SVI[30-55]	SVI > 38 mL/m^2 (low), <31 ml/m^2 (high)	Low vs high risk		
Svo_2[30,54,33,38]	>65% (low) | <60% (high)	Low vs high risk	Perfusion adequacy; correlates with outcome.	Sampling-site dependence.

Please note that the stated thresholds are derived from referenced clinical studies identifying adverse outcomes in persons with RHF.

AHF, acute heart failure; CI, cardiac index; CO, cardiac output; HF, heart failure; LVAD, left ventricular assist device; mPAP, mean pulmonary artery pressure; PADP, pulmonary artery diastolic pressure; PAH, pulmonary arterial hypertension; PAPi, pulmonary artery pulsatility index; PASP, pulmonary artery systolic pressure; PCWP, pulmonary capillary wedge pressure; PH, pulmonary hypertension; PVR, pulmonary vascular resistance; RAP, right atrial pressure; RHF, right-heart failure; RV, right ventricular; RVF, right ventricular failure; RV-MPS, right ventricular myocardial power-to-stroke volume ratio; RV−PA, right ventricular−pulmonary arterial; RVSWI, right ventricular stroke-work index; SVI, stroke-volume index; Svo_2, mixed venous oxygen saturation; TD, thermodilution; WU, Wood units.

vs right-sided pathology and must be interpreted in conjunction with other hemodynamic indices (eg, RA pressure), multimodal imaging, and clinical context. Among patients with adjudicated PAH from the Pulmonary Vascular Disease Phenomics (PVDOMICS) program, approximately 1 in 5 patients who were adjudicated had an elevation in resting PAWP.[58] Of note, these patients with group 1 PH were metabolomically and biologically distinct from combined pre- and postcapillary PH, emphasizing the limitations of using resting PAWP alone to separate the 2 groups. Hence, in situations in which PAWP is normal at rest but the clinical context suggests left heart disease, provocative maneuvers such as fluid or exercise challenge may be required to unmask occult left-sided dysfunction and clarify the driving pathophysiology. In contrast, in situations in which PAWP is elevated, PAWP saturation should be drawn to confirm complete occlusion and accuracy of the measurement.[59] Integration of these tools is critical to avoid misclassification, inappropriate therapy, and adverse outcomes.

In mild RHF, elevated filling pressures as a sign of right-sided congestion might be the only hemodynamic derangement. In other words, CO is still preserved. As RHF progresses, and RV systolic function fails to compensate for a dilated, dysfunctional RV, systemic CO will decrease.[26,30,55] In the acutely decompensated setting, in which time-dependent decisions regarding management need to be made, the pulmonary artery pulsatility index (PAPi, which is a ratio of PA pulse pressure/RAP) can be a useful tool.[60-63] As such, it reflects both the native contractility of the RV as well as venous congestion. A low PAPi would indicate reduced RV pulsatility and high levels of congestion. PAPi cutoffs as a marker of severe RV dysfunction have been validated in several clinical scenarios, including acute inferior myocardial infarction and post-LVAD (Table 4), and are used for clinical decision-making regarding escalation to tMCS.[64,65] In chronic disease states (heart failure or PH), average PAPi will be considerably greater than in the acutely decompensated phases, though it can still be useful as a predictor of long-term outcomes, such as death, need for heart transplantation, or rehospitalization.[50,66,67] PAPi is most useful in the setting of elevated RAP.

Pressure-volume (PV) loop analysis provides a mechanistic framework for understanding RV adaptation and failure. Coupling occurs when RV function is adapted to the pulmonary vascular load and is described by systolic and arterial elastance, E_{es}/E_a. A well-adapted RV often has preserved ventriculoarterial coupling, whereas maladapted RVs have varying degrees of altered ventriculoarterial coupling (Fig. 1) In the normal RV, contractility (Emax) is appropriately matched to PA afterload, resulting in efficient RV-PA coupling and preservation of stroke volume. In acute RV failure, such as with acute ischemia, contractility decreases and volume increases, without abrupt changes in afterload, resulting in a PV loop that is shifted

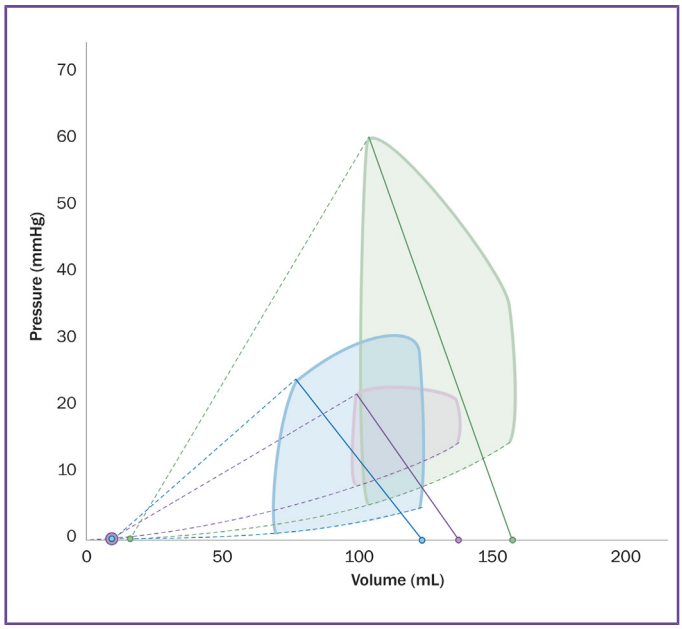


Fig 1. Pressure-volume (PV) loop analysis with a mechanistic framework for understanding right ventricular (RV) adaptation and failure. Blue/normal RV function: The loop displays a characteristic morphology with a rounded pressure increase and gradual pressure decay during ejection, reflecting the low-impedance pulmonary vascular bed. The steep Ees slope relative to Ea indicates preserved RV-pulmonary artery (PA) coupling efficiency. Purple/acute RV failure: An acute reduction in RV contractility results in a rightward shift of the loop, with increased end-diastolic and end-systolic volumes and reduced peak systolic pressure. The loop area is diminished, reflecting depressed stroke volume and stroke work. The flattened Ees slope reflects uncoupling of the RV from the pulmonary arterial system, primarily driven by reduced contractility, before compensatory remodeling can occur. Green: chronic pulmonary hypertension with RV adaptation. Sustained afterload elevation drives RV hypertrophy and remodeling, resulting in markedly increased peak systolic pressure and a steeply increasing end-systolic pressure-volume relationship. End-diastolic volume is increased with a preserved stroke volume. The loop adopts a more trapezoid shape with elevated both systolic and diastolic pressures, distinguishing this adapted state from acute decompensation. A midsystolic notch can also be seen in this phenotype, related to reduced PA compliance and wave reflections during systole.

to the right with reduced stroke volume, stroke work, and pressures. In contrast, as pulmonary vascular resistance (PVR) is chronically elevated (eg, in PAH), the RV initially augments contractility, shifting the end-systolic PV relationship upward and increasing peak systolic pressure with a preservered stroke volume. RV systolic pressures usually increase until end-systole (ie, trapezoid shaped) and may present with a midsystolic notch due to wave reflections in the setting of reduced PA compliance.[68]

Prognosis and Risk Stratification

The prognosis of RHF is heterogeneous and depends on etiology, chronicity, and severity of disease. Risk stratification requires an integrated assessment of RV structure and function together with systemic consequences of RV dysfunction.[69] Several validated scoring systems exist,

Table 5 Multiparametric risk-stratification models used in PAH

Comprehensive 3-strata model[30]	Low/intermediate/high	Stepwise increase in 1-y mortality: <5%, 5%–20%, >20%. Multiparametric risk stratification integrating clinical, functional, biochemical, imaging, and invasive parameters at baseline.	Based on pooled registry data (COMPERA, REVEAL, French, SPAHR). Core tool for initial diagnostic risk evaluation; used to guide therapy escalation.	Intermediate risk may include heterogeneous phenotypes; some parameters not universally available.
Simplified 4-strata[30]	Low/intermediate-low/intermediate-high/high	WHO-FC, 6MWD, BNP/NT-proBNP; points 1–4 → mean rounded to nearest integer. Guides treatment escalation.	Developed from COMPERA and SPAHR registries. Recommended for follow-up risk assessment.	Largely retrospective validation. Simpler but limited to 3 core variables; may underrepresent hemodynamic or imaging deterioration.
REVEAL 2.0[69]	Weighted 13-variable score (low ≤6; high ≥9)	Provides continuous, validated 1-y mortality prediction	Derived from 2529 patients with PAH in US REVEAL Registry; c-index ≈ 0.76; strongest prognostic accuracy among registry-based models.	More complex; needs full data set.
REVEAL Lite 2[8]	Six-variable noninvasive score Low: ≤5, high: ≥8	Provides continuous, validated 1-y mortality prediction	Simplified tool with good discrimination. Validated in REVEAL Registry and real-world cohorts.	Excludes invasive data; may underperform in specific phenotypes.
REVEAL-Echo[21]	Echo-enhanced model	Risk score adjunct	Improved discrimination vs REVEAL Lite.	Needs quality echo data.

6MWD, 6-minute walk distance; BNP, B-type natriuretic peptide; COMPERA, Comparative, Prospective Registry of Newly Initiated Therapies for Pulmonary Hypertension; ECHO, echocardiography; NT-proBNP, N-terminal pro–B-type natriuretic peptide; PAH, pulmonary arterial hypertension; REVEAL, Registry to Evaluate Early and Long-Term Pulmonary Arterial Hypertension Disease Management; REVEAL-ECHO, echocardiography-enhanced REVEAL risk model; SPAHR, Swedish Pulmonary Arterial Hypertension Registry; WHO-FC, World Health Organization functional class.

although most are context-specific (Table 5).[8,21,30,70] In PAH, multiparametric approaches such as REVEAL 2.0 and the simplified REVEAL Lite incorporate clinical, biomarker, imaging, and hemodynamic data to estimate 1-year mortality and are used in practice to guide treatment escalation, including the initiation of parenteral prostacyclin therapy or consideration of transplant referral.[8,71] The 2022 European Society of Cardiology/European Respiratory Society guidelines similarly endorse a "traffic light" model based on World Health Organization functional class, 6-minute walk distance, natriuretic peptides, imaging, and invasive hemodynamics to estimate prognosis and direct the intensity of therapy.[30] In advanced HF populations undergoing LVAD implantation, tools such as the European Registry for Patients with Mechanical Circulatory Support (EUROMACS)-RHF score, STOP-RVF score, and others (eg, Penn, Kormos, Michigan, CRITT) have been developed to predict postoperative RHF; in this setting, they are applied to inform patient selection, risk counseling, perioperative planning, and the need for biventricular support.[8,21,53,70,72-79] These scores integrate clinical and hemodynamic parameters with only modest performance and remain largely confined to device-related contexts in the absence of broad validation in nondevice RHF population. Beyond formal scores, RV-focused hemodynamics, both at rest and with fluid, drug, ventilatory, or exercise perturbations, as well as markers of end-organ function inform RV risk.[80-85] Taken together, existing evidence supports a multiparametric approach that combines clinical status, biomarkers, imaging, and invasive hemodynamics rather than reliance on any single variable or score.

Management of RHF

The management of RHF requires a pathophysiology-driven approach that targets the underlying etiology of RHF, as well as its chronicity and comorbidities. For acute RHF, therapeutic decision-making may involve decongestion, vasopressor/inotrope use, or tMCS in the setting of RHF refractory to medical management. The 3 main tenets of a successful RHF management strategy include optimizing preload, reducing RV afterload, and augmenting contractility. Additional considerations for ensuring rate and rhythm control, restoring coronary blood flow, and maintaining adequate ventilatory and oxygenation thresholds can be considered for cases in which they apply.

Preload Optimization

In RHF, preload must be optimized according to the underlying disease state. In rare circumstances, intravenous (IV) fluid administration might be necessary to maintain adequate RV filling and stroke volume, such as in the setting of RV infarction and relative volume depletion. It is therefore critical to perform a proper volume assessment to assist with clinical decision-

making. Weighing the competing interests of both decongestion and perfusion is essential, because overdiuresis may lead to hypotension and renal dysfunction, especially in low-output RHF. In cases in which volume status remains unclear, renal function is worsening, or hemodynamic instability, invasive hemodynamics (eg, RAP, RAP/PAWP ratio) should be considered. Although previous studies have revealed that 70%–80% of patients will have concordant RAP and PAWP, there are cases in which discordant pressures may exist, such as in isolated RHF (high RAP, low/normal PAWP).[86-88] In this instance, studies have shown it is both safe and beneficial to diurese these patients.[89,90]

Diuretics are a Class I recommendation for treatment of patients with PAH and evidence of RHF with fluid retention for symptomatic relief and optimization of RV performance—both in relation to RV wall tension and the effect on the LV from associated interventricular dependence.[30,91] Loop diuretics continue to be the mainstay of therapy because of their rapid onset of action, potency, and efficacy.[92] Serial monitoring of kidney function is needed as these patients are at risk for developing acute kidney injury from both under- and overdiuresis. However, between 20%–50% of patients with HF will develop diuretic resistance, a complex and clinically challenging condition secondary to multiple mechanisms.[93,94] When encountered, combination diuretic therapy with both loop and thiazide diuretics may help combat diuretic resistance and increase sodium excretion.[95] However, persistent congestion, particularly in the face of worsening renal dysfunction, might precipitate more invasive maneuvers to remove volume, particularly ultrafiltration.[96-102]

Afterload Reduction

Pulmonary vasodilators are a mainstay of therapy for the management of select patients with acute RHF. These provide benefits by lowering PVR and PA pressure and improving oxygenation. Treatment options include IV and inhaled prostacyclin analogues (PCAs) and inhaled nitric oxide (iNO).[1] Inhaled options provide greater pulmonary selectivity with minimal to no systemic exposure, whereas IV therapies carry a greater risk of systemic adverse effects (ie, hypotension, ventilation/perfusion mismatch, and volume overload in certain patients).[1,103] Combination therapies that include a PCA are particularly effective by significantly lowering RV afterload for patients in whom RHF is the initial manifestation of PAH.[104] In addition, those with severe, refractory postoperative PAH or overt RHF may benefit from combination inhaled therapies.[103]

iNO has historically been the most commonly used vasodilator; however, the high cost and delivery system are disadvantages.[105] Short-term iNO lowered PVR and improved RV mechanics (increased RV ejection, attenuated RV distention) in patients with PH, RV myocardial infarction, pulmonary embolism, and after LVAD.[1] Doses ranged from 10 to 40 ppm, but exceeding 10 ppm was not associated with greater PVR reductions.[106] Greater doses may also be associated with adverse effects such as methemoglobinemia and increased bleeding time.[105] Weaning iNO is important to avoid the potential for rebound PAH.[107]

Epoprostenol (EPO) is a PCA that is generally preferred to be given IV for severe RV failure, particularly in the setting of PAH because of its potency, rapid onset, and ability to titrate to effect rapidly (~3- to 6-minute half-life).[108] EPO, both inhaled and administered IV, has been studied across a wide range of doses (5–120 ng · kg^{-1} · min^{-1}); however, adverse effects are prevalent and include hypotension, flushing, jaw pain, and gastrointestinal effects such as nausea and diarrhea. Current guidelines and consensus statements recognize inhaled EPO as a reasonable alternative to iNO for the management of postoperative PH and RV dysfunction after cardiac surgery, particularly in settings where iNO is not available or cost-prohibitive. Of note, inhaled EPO requires a specialized delivery system for administration.[109] Other less-validated therapies include inhaled iloprost, a PCA with a longer half-life (20–30 minutes) and inhaled milrinone.[103]

Contractility: Inotropic Management

Inotropic vasodilators are well-suited to treat patients with RVF on the basis of their pharmacologic effects of increasing myocardial contractility and lowering vascular resistance and filling pressures[110] (Fig. 2). The most commonly

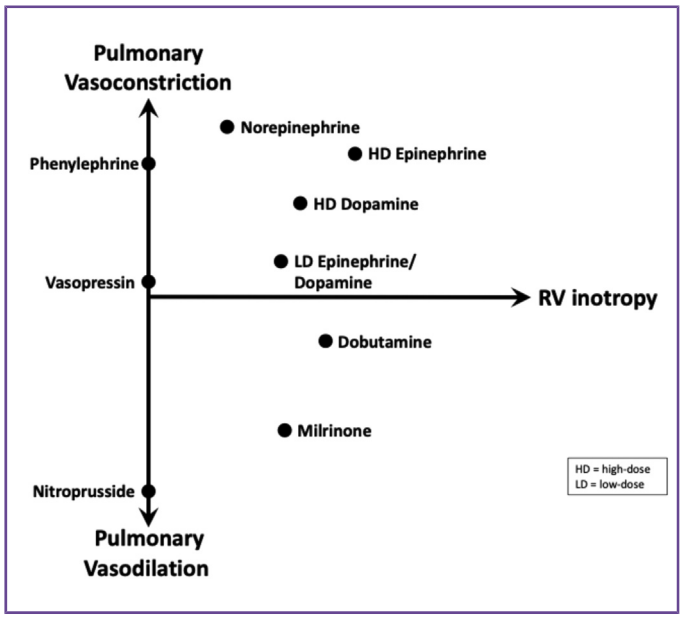


Fig 2. Vasoactive agents and effects on right ventricular (RV) dysfunction. HD, high-dose; LD, low-dose.

used agents are dobutamine and milrinone, which provide favorable hemodynamic improvements in RVF.[108] Dobutamine is a beta-one and beta-two agonist, whereas milrinone is a phosphodiesterase-type 3 inhibitor that augments downstream beta-adrenergic receptor signaling and activation.[111] Direct comparisons are limited but show similar clinical outcomes and adverse events.[1,112] Therefore, decisions to select a particular medication can be based on pharmacology, pharmacokinetics, and clinician experience.

Dobutamine is often used initially in patients who are acutely unstable because of its rapid onset and short half-life (~2 minutes), leading to prompt increases in CO. It also may be more useful among those with a lower baseline systolic BP (ie, <100 mm Hg) as it would be expected to cause less vasodilation than milrinone.[111] Typical doses are 2.5−7.5 $\mu g \cdot kg^{-1} \cdot min^{-1}$. Dopamine is an alternative to dobutamine but is limited by more modest inotropic effects and more prevalent tachyarrhythmias.[92] In comparison, milrinone exhibits more potent pulmonary and systemic vasodilatory effects, which can be advantageous in those with RVF due to PH.[113] Milrinone is also more useful for patients who recently received a beta-blocker due to impaired response with receptor antagonism.[114] Milrinone has a slower onset and longer half-life (~2−3 hours with normal kidney function) and progressive accumulation in those with kidney impairment. Typical doses are 0.25−0.5 $\mu g \cdot kg^{-1} \cdot min^{-1}$, but adjustments should be made for estimated glomerular filtration rate or creatinine clearance.[111] Disadvantages of both agents include the risks of arrhythmias, myocardial ischemia, and hypotension.

To maintain perfusion in patients who have profound hypotension and shock, an inotropic vasopressor may be required.[1,111] A MAP target of 70−75 mm Hg has been suggested to maintain aortic pressure, preserve right coronary perfusion, and mitigate RV ischemia, although no studies have directly compared alternative MAP targets in RHF.[92] Low-dose norepinephrine and vasopressin are preferred vasopressor options.[92,108] The use of norepinephrine decreased the ratio of PAP to systolic blood pressure without a change in cardiac index due to its balanced alpha- and beta-effects.[105] Vasopressin, unlike catecholamines, has a pulmonary vasodilator effect, natriuretic effect, and a lower risk of tachyarrhythmias.[92,115] Epinephrine is not preferred because of its potent alpha- and beta-effects, leading to lactic acidosis and tachycardia, but may be required in refractory cases with severely reduced CO.[92,104]

Although digoxin is often used in patients with PH-related acute RHF, given its acute increase in CO, this practice is largely based on limited preclinical and acute clinical hemodynamic studies.[116,117] In a large, retrospective, nonrandomized cohort, chronic use of digoxin in patients with PAH was associated with greater all−cause mortality and HF hospitalization, even after multivariate correction.[118]

Chronotropy: Rhythm Management

Maintaining optimal heart rate and rhythm control is essential for RV function. Both intact atrial contraction ("atrial kick") and atrioventricular synchrony contribute significantly to RV filling, which can be disrupted in the setting of supraventricular tachyarrhythmias. Supraventricular tachyarrhythmias can lead to reductions in ventricular filling and thus right-sided CO, which may further exacerbate RHF. Although specific rate control agents, such as beta-blockers or nondihydropyridine calcium channel blockers, may be initially effective, their use should be heavily weighed against their negative inotropic effects, which may potentiate hemodynamic decompensation, as well as the risks of systemic hypotension.[119] In addition, beta-blockade might also contribute to other mechanisms leading to worsening RHF, including bronchoconstriction and hypoxia, pulmonary vasoconstriction, and increased RV afterload. Direct current cardioversion may be attempted, although success rates are variable in critical illness.[120-123] Antiarrhythmic agents should be used with caution, given their negative chronotropic effects.

Pulmonary/Critical Care Aspects

Respiratory management plays a central role in acute RHF, because of limited baseline cardiopulmonary and hemodynamic susceptibility to both anesthetic agents and positive-pressure ventilation.[124,125] Positive end-expiratory pressure is known to decrease both RV and LV preload.[126,127] Although decreased RV preload in the setting of RHF may be beneficial, the RV Frank-Starling curve can be unpredictable and decreased LV preload in a patient with impaired RV to LV transit (from PH or RHF) may have catastrophic hemodynamic consequences. Beyond the effects on venous return, high airway pressures and lung overdistention also lead to increased pulmonary vascular resistance by compression of the alveolar capillary unit, thereby increasing RV afterload and worsening RV-pulmonary arterial coupling.[127] For this reason, high positive end-expiratory pressure should be avoided and low tidal volume ventilation used to reduce intrathoracic pressures.[128,129]

Mechanical ventilation should be used judiciously, as noninvasive ventilation is preferred when feasible.[130] Anesthetics used for induction and sedatives may precipitate systemic hypotension through sympatholytic effects, thereby reducing coronary perfusion and RV preload. Hypoxia, hypercapnia, and acidosis all may worsen pulmonary vasoconstriction, contributing to greater RV afterload and thus worsening RHF and risk of peri-intubation cardiac arrest. Use of ketamine

during emergent intubation has been associated with a greater risk of cardiovascular derangements in the setting of RVF, but etomidate has a lesser direct myocardial depressant effect.[131] Hence, awake tracheal intubation is preferred over rapid sequence intubation for patients with RVF, when feasible.[132]

Role of tMCS Devices

tMCS devices play an increasingly pivotal role in managing CS with acute RHF by relieving venous congestion and augmenting CO.[133] These devices may serve as a bridge to recovery, decision-making, durable support, or heart transplantation. Available options, including Impella RP and RP Flex (J&J MedTech, New Brunswick, NJ), ProtekDuo (LivaNova, Houston, TX), surgical right ventricular assist devices (RVADs), and central or peripheral ventriculoarterial extracorporeal membrane oxygenation (VA-ECMO), provide varying degrees of circulatory and oxygenation support (Table 6, Fig. 3) Optimal use requires careful patient and device selection on the basis of the etiology of RHF, need for isolated vs biventricular support, hemodynamic status, anatomic considerations, end-organ

Table 6 Temporary MCS devices in RHF management

Device Type	Indications	Hemodynamic Targets	Need for Escalation	De-escalation	Weaning Steps	Explant Notes
Impella RP (direct IVC → PA bypass; femoral venous introducer sheath 23 Fr)	Isolated RV failure with cardiogenic shock when rapid catheter-laboratory deployment is needed; no major oxygenation deficit; no profound shock; short duration of support; ambulation not feasible.	Flow >4 L/min achievable; heparinized purge + systemic AC typically used (consider bicarbonate purge if bleeding risk).	Indications: rising lactate, persistent RA/CVP elevation, CI <2.2 L· min^{-1} · $m^{2\ -1}$ despite max support, hypoxemia emerges, device malposition/hemolysis. Actions: confirm position; increase speed; add pulmonary vasodilator; if oxygenation or biventricular failure → convert to ProtekDuo (± oxygenator) and LVAD or VA-ECMO.	Euvolemia (RA ≤12 mmHg), CI ≥2.2 L · min^{-1} · m^{2-1}, PA sat ≥50%, PAPi >1.0, lactate <2 mmol/L, RV function improvement on TEE/TTE, end-organ function improvement.	Reduce flow by 0.5–1.0 L/min every 2–3 h with hemodynamic and echo monitoring. Maintain flow ~2 L/min for 6–8 h before device removal. Consider inotropic up-titration and pulmonary vasodilators. If instability → abort and re-optimize.	Hold/minimize systemic AC; may use manual pressure, suture closure or percutaneous closure device after catheter/sheath removal.
Impella RP Flex (direct IJ/RA→PA; internal jugular introducer sheath 23 Fr)	As above plus: anticipated >24–48 h support and desire for mobilization (IJ access).	Similar to RP; easier to mobilize; manufacturer recommends purge + systemic AC.	As above, favored step-up from femoral RP if prolonged course or to enable PT.	As above.	As above (stepwise ↓ flow), often easier to tolerate due to stable IJ position.	Bedside venous sheath/catheter removal from IJ with skin suture closure of the insertion site.
ProtekDuo cannula + extracorporeal pump (direct RA→PA via IJ 29–31 Fr; subclavian vein is an alternative; oxygenator optional)	RV failure with significant hypoxemia or need for longer support/ambulation; rapid catheter laboratory or operating room placement; configurable with oxygenator (veno-pulmonary arterial ECMO).	Flows up to 4 L/min for 29 Fr cannula and up to 5 L/min for 31 Fr cannula; systemic AC required.	Indications: inadequate flow (high PA afterload), hypoxemia, hemolysis, SVC syndrome, significant PI. Actions: add/integrate oxygenator in patients with hypoxemia; upsize/optimize cannula position; if biventricular or end-organ failure → convert to VA-ECMO.	As above; ensure adequate lung gas exchange if oxygenator used (Pao_2/Fio_2 improving, Fio_2 ≤0.5, PEEP <10).	Stepwise pump-flow reduction with echo/hemodynamic targets as above; watch RA pressure, PAPi, MAP; ensure stable oxygenation/ventilation when oxygenator is weaned.	Bedside/operating room decannulation with skin suture closure of the insertion site (large-bore cannula).
Surgical RVAD (eg, CentriMag) – RA (or femoral vein and PA cannulation	Postcardiotomy RV failure; severe RV failure during durable LVAD implantation; need for high, durable flows; easy integration with oxygenator; ambulatory	Up to ~10 L/min, depending on venous cannula size; systemic AC required.	Indications: persistent venous congestion/low CO, high afterload limiting flow, circuit issues. Actions: add oxygenator; adjust cannula size/position; consider transition to VA-ECMO or durable strategy (BiVAD/TAH/	Same hemodynamic and perfusion targets as above; ensure oxygenation /ventilation remain adequate if oxygenator used.	Gradual RPM/flow down-titration with invasive monitoring as above. Ensure adequate lung function if oxygenator is used.	Operating room removal for direct RA/PA cannulas, which requires re-sternotomy; bedside noninvasive decannulation is possible when

(*Continued*)

Table 6 (*Continued*)

Device Type	Indications	Hemodynamic Targets	Need for Escalation	De-escalation	Weaning Steps	Explant Notes
	potential with sternum closed.		transplant) if recovery unlikely.			cannulated PA tunneled graft and percutaneous femoral venous cannulation approach is used.
VA-ECMO (indirect: RA/IVC drainage → arterial return; femoral or central aortic)	Profound RV failure leading to shock /arrest, allows rapid peripheral initiation (requires cannulation expertise), biventricular failure, severe hypoxemia /hypercapnia; ambulation usually restricted with femoral arterial cannulation (but possible).	Robust arterial flow to restores systemic perfusion; anticipate ↑ LV afterload (plan LV venting if needed); AC: heparin commonly used (growing use of bivalirudin and argatroban).	Indications/actions: if insufficient flows or persistent end-organ hypoperfusion consider upsizing cannulae; if LV distention/ severe pulmonary edema add LV venting strategy (IABP, Impella, LA vent, etc), if persistent hypoxemia convert to VAV-ECMO.	Ramp down trial goals: MAP ≥60 mm Hg, RA ≤12 mm Hg, PAWP ≤18 mm Hg, PAPi >1.0, $CV-O_2$ ≥60%, improving oxygenation; low inotropic/vasopressor support; echocardiographic signs of RV recovery.	Stepwise ↓ flows to ~2 L/min for ~6 h while maintaining AC; consider "bridge trial" (inflow−outflow bridge) to observe completely off-flow; ensure native lungs carry ≥50%−80% gas exchange (Pao_2/Fio_2 >100 on low-moderate settings).	Central cannulation: operating room removal. Peripheral: cutdown with vessel repair (operating room) or percutaneous closure (bedside/catheter laboratory/ operating room). Monitor for limb ischemia and bleeding complications

AC, anticoagulant; BiVAD, biventricular assist device; CI, cardiac index; CVP, central venous pressure; Fio_2, fraction of inspired oxygen; IJ, internal jugular; IVC, inferior vena cava; LV, left ventricular; LVAD, left ventricular assist device; PA, pulmonary artery; Pao_2, partial pressure of oxygen; PAPi, pulmonary artery pressure index; MAP, mean arterial pressure; PEEP, peak end-expiratory pressure; PI, pulmonary insufficiency; PT, physical therapy; RA, right arterial; RV, right ventricular; RVAD, right ventricular assist device; SVC, superior vena cava; TAH, total artificial heart; TEE, transesophageal echocardiogram; TTE, transthoracic echocardiography; VA-ECMO, VA-ECMO, ventriculoarterial extracorporeal membrane oxygenation; VAV-ECMO, ventriculoarterial venous extracorporeal membrane oxygenation.

function, ambulation potential, and institutional expertise. Management during tMCS necessitates a multidisciplinary approach aimed at optimizing hemodynamics, oxygen delivery, and end-organ perfusion while balancing anticoagulation and bleeding risk.[134] Early mobilization, nutritional support, and vigilant monitoring for complications are essential. Regular reassessment of myocardial recovery, neurologic function, and multiorgan status informs ongoing therapy and decisions regarding escalation, transition, or withdrawal of support. Weaning should begin once the primary insult has resolved and RV recovery is evident. A structured, closely monitored reduction in device output, guided by clinical, hemodynamic, and echocardiographic parameters, ensures safe discontinuation, while transient inotrope support may facilitate successful explantation (Fig. 4)

Role of Transplantation

Given the significant limitations of durable MCS technology for patients with isolated RHF or biventricular failure, heart transplantation remains the definitive heart-replacement therapy in these scenarios. Transplantation is an important option in patients with chronic or irreversible RHF that is unresponsive to pharmacologic therapy, including patients with long-standing biventricular failure, congenital heart disease with RV dominance, severe right-sided valvular disease, arrhythmogenic right ventricular cardiomyopathy, etc. In addition, patients supported with a durable LVAD who develop progressive RVF and become dependent on inotropes or temporary RVADs should be evaluated for heart transplantation.[135] In patients with advanced PH, determining whether to pursue heart, lung, or combined heart-lung transplantation is a complex decision, often influenced by the presence and extent of Eisenmenger physiology and RV contractile reserve.

Supportive/Palliative Care

Limited invasive palliative options exist for severe RHF. Balloon atrial septostomy is a transcatheter technique that creates a right-to-left shunt across the interatrial septum as a mechanism to decompress the right heart and improve CO, particularly for patients with end-stage PH. The procedure was initially introduced decades ago for patients with congenital heart disease.[136] but has been used more recently for symptom

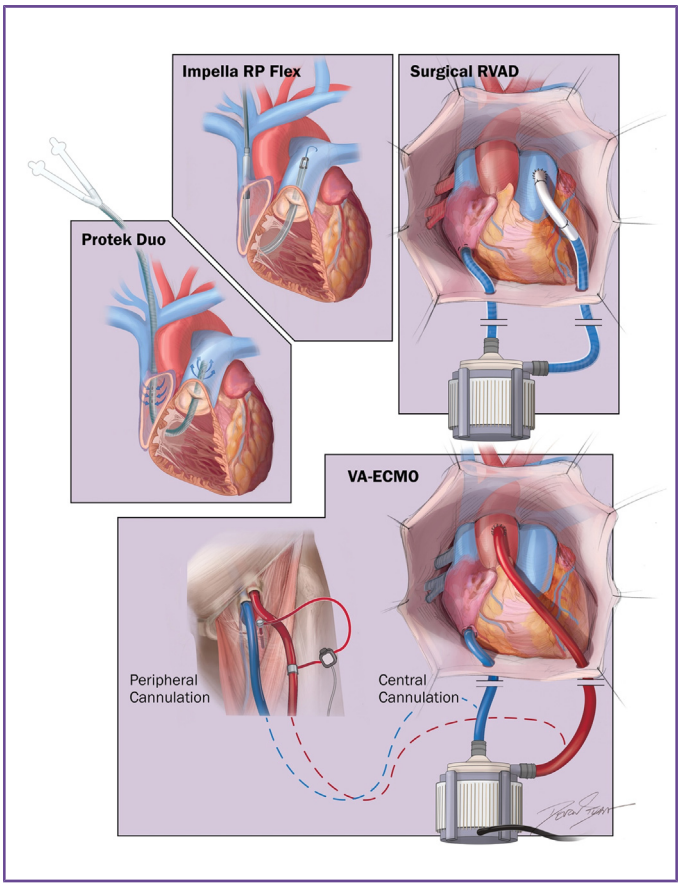


Fig 3. Temporary mechanical circulatory support devices in right heart failure. RVAD, right ventricular assist device, VA-ECMO, ventriculoarterial extracorporeal membrane oxygenation. Figure adapted with permission from J Heart Lung Transplant 2024;43:1894–904.

management or as a bridge until lung transplantation.[137] However, balloon septostomy is not an option for all patients, particularly those with elevated RAP >20 mm Hg, significant hypoxia, or an elevated PVR index >55 U/m^2.[138]

RHF in specific situations

Acute Pulmonary Embolism (PE)

RHF occurring in the setting of acute PE represents an increasingly complex diagnostic and nuanced treatment challenge. PE only results in elevated PA pressures once at least 25%–30% of the pulmonary vasculature is occluded by clot, in the absence of pre-existing cardiopulmonary disease.[139] As clot burden increases, acute RHF will develop once PA pressures exceed 40 mm Hg, leading to RV dilatation, reduced LV preload and hence coronary perfusion, and eventually shock.[139] Therefore, current clinical practice guidelines recommend performing risk stratification for mortality, categorizing patients into 4 groups (low, low-intermediate, intermediate-high, and high) using a combination of clinical symptoms and hemodynamic instability together with laboratory and imaging indicators of RV dysfunction.[140,141] Patients who have some degree of RHF will fall in the intermediate- to high-risk category, for whom there are a multitude of non-MCS therapeutic options with varying risks and benefits. Depending on the extent and pattern of clot burden, as well as the severity of hypoxia and hemodynamic derangement, PE should be addressed with appropriate

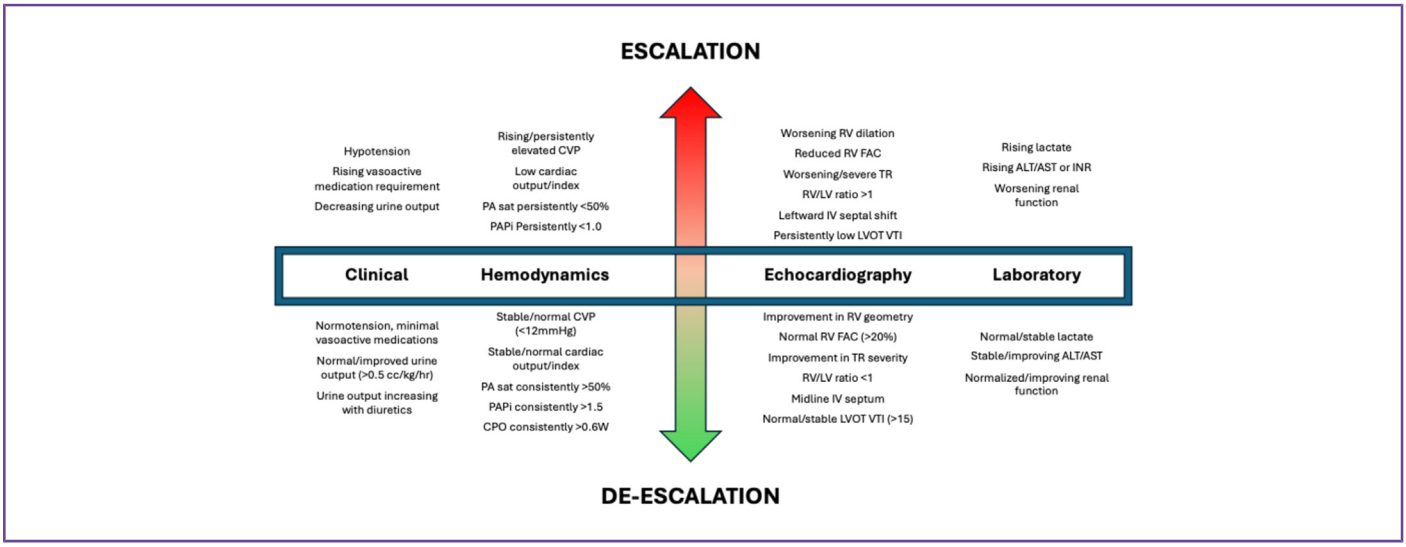


Fig 4. Considerations for escalation and de-escalation of temporary circulatory support devices in RHF. ALT, alanine transaminase; AST, aspartate transaminase; CPO, cardiac power output; CVP, central venous pressure; FAC, fractional area change; INR, international normalized ratio; LVOT, left ventricular outflow tract; PA, pulmonary artery; PAPi, pulmonary artery pulsatility index; RHF, right heart failure; RV, right ventricle; TR, tricuspid regurgitation; VTI, velocity time integral.

intervention, such as thrombectomy, catheter-guided thrombolytics, or consideration of VA-ECMO in select cases.[142-145] Although the strengths of the recommendation vary, most guidelines advocate for a multidisciplinary PA response team to facilitate risk stratification and expedite decision-making to effectively manage patients with PE and any degree of RVF.[140]

Perioperative Setting

Primary isolated early RHF after orthotopic heart transplantation occurs in up to 20% of cases and is associated with poor outcomes. The pathogenesis of RHF is multifactorial and complex, and it may include ischemia-reperfusion injury, elevated PVR, inadequate donor-recipient size matching, and other poorly understood pathophysiologic processes. The diagnosis of severe primary graft dysfunction-RV after heart transplantation is based on abnormal hemodynamic parameters—including a RAP >15 mm Hg, PAWP <15 mm Hg, cardiac index <2.0 L/min/m^2, a transpulmonary gradient (TPG) <15, and PA systolic pressure <50 mm Hg—or the need for temporary MCS including VA-ECMO or RVAD support.[146] Pharmacologic management of RHF and PH after heart transplantation should focus on optimizing preload and contractility while reducing PVR.[147] In severe cases of primary graft dysfunction, tMCS such as VA-ECMO or temporary RVADs may be required. Meticulous donor selection, preoperative optimization of PVR, vigilant hemodynamic monitoring, and excellent allograft-preservation techniques are key preventive strategies.

Acute RHF after LVAD implantation occurs in up to 1 in 4 patients, necessitating prolonged use of vasoactive medications or use of tMCS for RV support. It is often precipitated by intraoperative factors that acutely alter RV loading, geometry, and perfusion. LVAD activation causes immediate leftward septal shift and ventricular remodeling, impairing RV contractility through increased free-wall stress and loss of interventricular dependence. Intraoperative air embolism during deairing may further compromise RV function via transient right coronary ischemia. Intraoperative bleeding, transfusion-related volume overload, and mediastinal tamponade can further exacerbate RV dysfunction, contributing to early postoperative RHF. Concomitant mitral valve surgery increases risk of left circumflex coronary compromise, leading to ischemia and adverse septal mechanics. In addition, general anesthesia and positive-pressure ventilation reduce venous return, increase intrathoracic pressure, and raise pulmonary vascular resistance, collectively increasing RV afterload.

Early RHF is associated with a significantly greater morbidity and mortality.[148] Current practices focus on the aforementioned diagnosis and management of RHF; however, there is no consensus strategy for management in this population.[148] A standardized protocol optimizing the tenants of acute RHF management—optimized RV preload, perfusion, contractility, and geometry; RV afterload reduction; careful ventilatory management; and rate and rhythm control—was found to reduce the rate of early RHF in patients after durable LVAD implantation.[149] In patients with LVADs and RHF, phosphodiesterase-5 inhibitors have resulted in mixed results—namely a reduction in PVR with improvement in RV-PA coupling but without clear long-term effects on RHF and increase bleeding risk.[150-152] Meanwhile, in the randomized controlled trial SOPRANO, macitentan modestly reduced PVR but without associated improvements in right heart hemodynamics, New York Heart Association functional class, or natriuretic peptides in early postoperative patients after LVAD implantation.[153] Routine use of these therapies in patients with LVADs is not recommended at this time, and additional research is needed.

Postoperative RHF can also occur after non−HF cardiac surgical interventions, particularly after procedures involving valvular surgery, congenital heart disease repair, coronary artery bypass grafting, or lung transplantation.[154] Mechanisms are multifactorial and include perioperative ischemic injury to the RV, air embolism to the right coronary artery prolonged cardiopulmonary bypass time, acute increases in preload, residual PH, and arrhythmias. Management focuses on adequate oxygenation/ventilation, sedation, acid-base equilibrium, and perfusion, with the goal to prevent a disproportionate increase in the RV afterload.

Pulmonary Arterial Hypertension

Pulmonary vasodilator therapies should be prescribed in patients with PAH with RHF according to the most recent guidelines.[30,155] Management of chronic RHF in PAH focuses on reducing RV afterload, optimizing preload, and supporting RV function (as discussed previously). Targeted pulmonary vasodilator therapies—including endothelin receptor antagonists, phosphodiesterase-5 inhibitors, soluble guanylate cyclase stimulators, and prostacyclin pathway agents—as well as the activin signaling inhibitor sotatercept—are central to treatment and aim to lower PVR and improve RV−pulmonary artery coupling. In advanced disease, escalation to parenteral prostacyclin therapy, referral to specialized PH centers, and consideration of advanced therapies such as lung or heart-lung transplantation may be necessary.

Left Heart Disease

Routine use of pulmonary vasodilators to reduce afterload in patients with PH associated with left heart disease, which includes both isolated and combined pre- and postcapillary PH, is not currently recommended, attributed to a combination of lack of benefit and risk of harm as the

result of fluid retention.[156] Patients with chronic RHF attributable to heart failure with reduced ejection fraction (HFrEF) and heart failure with preserved ejection fraction (HFpEF) should be managed according to the current guidelines for management of HF.[157,158] The CAMEO-DAPA (Evaluation of the Cardiac and Metabolic Effects of Dapagliflozin in Heart Failure With Preserved Ejection Fraction) trial suggested dapagliflozin can reduce PA, wedge, and RAP, although it was not specifically designed for RHF.[159] Sodium-glucose cotransporter-2 inhibitors also have been shown to improve echocardiographic measures of RV function (eg, TAPSE, systolic PAP, RV S') and lower pulmonary pressures, although again not specifically in patients with RHF.[160,161] Therapies studied in HFpEF with or without RHF (eg, macitentan, tadalafil) have not shown benefit and may cause harm.[162] Studies repurposing PAH therapy to patients with HFrEF-PH are generally small and associated with mixed results. The VICTORIA (Vericiguat Global Study in Subjects with Heart Failure with Reduced Ejection Fraction) trial randomized patients with HFrEF to receive the soluble guanylyl cyclase stimulator vericiguat or placebo in addition to guideline-based medical therapy.[163] However, clinical data on the PH profile of the study patients was not captured, thereby limiting insight into any potential benefit of RHF. The recently published phase 2 trial CADENCE provided proof of concept for improved pulmonary vascular and cardiac hemodynamics after activin signaling inhibition with sotatercept in patients with combined pre- and postcapillary PH in HFpEF.[164]

It is important to note that PAH-targeted therapies have reduced efficacy and tolerability and a greater chance of treatment failure among patients with PAH and multiple left heart disease risk factors. Such patients with overlapping phenotypes are prevalent and evidence-based treatment strategies are currently lacking. Therefore, patients should be considered for clinical trial enrollment rather than for treatment with PAH-specific therapies in these circumstances.[156]

Chronic Thromboembolic PH

Pulmonary thromboendarterectomy is the gold-standard, guideline-recommended treatment of choice that leads to normalization of pulmonary artery pressures in 75% of patients with a survival rate of >90% at 3 years.[165-167] Balloon pulmonary angioplasty has now become a guideline-recommended treatment for patients with persistent or recurrent PH after surgery, for those with distal disease not amenable to surgery, or if they are at prohibitive risk for surgery.[168] Lastly, there is increasing evidence from randomized controlled trials supporting the benefits of PAH medical therapy, which has informed the understanding of the efficacy and timing of treatments.[169] Similar to acute PE, the evolving treatment options and combinations of therapies have underscored the importance of a multidisciplinary team in comprehensively managing chronic thromboembolic PH.[165]

Valvular Diseases

Among adults older than the age of 65 years, 7.2% were found to have moderate-to-severe TR in a community-based prospective study.[170] TR is a heterogeneous disease, with determination of the causative disease process (primary, secondary, or cardiac implantable electronic device–related) critical to determining the appropriate treatment pathways.[171] Regardless of etiology, once TR develops, there is an association with increased mortality.[172-174]

For patients with symptomatic severe TR after optimal medical therapy targeting the underlying cause, the recommended intervention will depend on the degree of RHF, in addition to symptoms and comorbidities that impact surgical risk and whether valve anatomy is suitable for devices. Treatment options include surgical repair and replacement, as well as transcatheter interventions, including transcatheter repair and valve-in-valve implantation. Surgery for isolated TR is controversial, although it is generally preferred in patients with lower risk before the development of significant RVF. Current trials have demonstrated improvements in quality of life with transcatheter interventions. In the TRILUMINATE (Trial to Evaluate Cardiovascular Outcomes in Patients Treated with the Tricuspid Valve Repair System Pivotal) randomized controlled trial, tricuspid transcatheter edge-to-edge repair was found to be safe with a 30-day freedom from adverse events of 1.7%, reducing TR, and improving quality of life by Kansas City Cardiomyopathy Questionnaire compared with medical therapy.[175] Similarly, transcatheter tricuspid-valve replacement was found to improve HF symptoms and quality of life compared with medical treatment in the TRISCEND II (EVOQUE Transcatheter Tricuspid Valve Replacement: Pivotal Clinical Investigation of Safety and Clinical Efficacy Using a Novel Device) trial.[176] Whether transcatheter interventions improve outcomes, including mortality, with longer-term follow-up, remains to be seen.

Left-sided valvular heart disease (mitral regurgitation or stenosis and advanced aortic stenosis) is another common cause of chronic RHF through the development of pulmonary venous hypertension, which increases RV afterload and secondary pulmonary vessel remodeling. Over time, the resulting PH contributes to progressive RV dilatation and systolic dysfunction, often accompanied by secondary TR. Management focuses on timely treatment of the underlying valvular lesion and optimization of volume status to prevent irreversible RV failure.

Others

Chronic RV pacing can adversely affect right heart structure and function through mechanisms related to

electrical and mechanical dyssynchrony, which may impair RV contractility and contribute to progressive RV dysfunction. In addition, transtricuspid pacing leads may interfere with tricuspid valve leaflet motion, resulting in lead-associated TR and subsequent right atrial and ventricular dilatation. In patients with a high RV pacing burden and ventricular dysfunction, upgrading to cardiac resynchronization therapy can restore more coordinated ventricular activation and improve (both) ventricular performances. Increasingly, conduction system pacing—such as His bundle pacing or left bundle branch area pacing—may be used to achieve more physiologic activation and avoid further dyssynchrony. If device-related TR is present, lead revision or extraction in selected cases may be required.

Health system and economic impact

Although RHF represents a major source of morbidity and health care use across diverse cardiovascular populations, its system-wide and economic consequences remain underappreciated. In PH, RHF represents the main driver of acute decompensation. Patients with RHF experience longer hospitalizations, greater rates of intensive care unit admission, and a greater need for invasive monitoring and circulatory support than those with isolated left-sided failure. The frequent procedural burden (need for right heart catheterization, dialysis, and MCS) and considerations for advanced therapies, further contribute to straining intensive care unit resources. Similar to LHF, readmission rates remain high. Beyond hospital-based costs, RHF contributes to lost productivity, caregiver strain, and diminished quality of life, particularly in younger patients with pulmonary vascular disease.

Despite consuming significant health care resources, therapeutic options remain limited, and current reimbursement structures do not adequately capture the complexity or intensity of care required. This mismatch underscores the need for dedicated RHF care pathways and more precise coding strategies to improve both resource allocation and epidemiologic tracking. Furthermore, improved documentation and health system awareness could enhance reimbursement accuracy and stimulate investment in targeted innovation.

Current gaps and future directions

The persistent lack of progress in RHF research reflects several fundamental and interrelated gaps, as illustrated in Figure 5. First, the absence of a standardized, universally accepted definition of RHF across phenotypes and care settings has impeded trial design, data harmonization, and therapeutic development. Current criteria are subjective and vary across disease contexts,

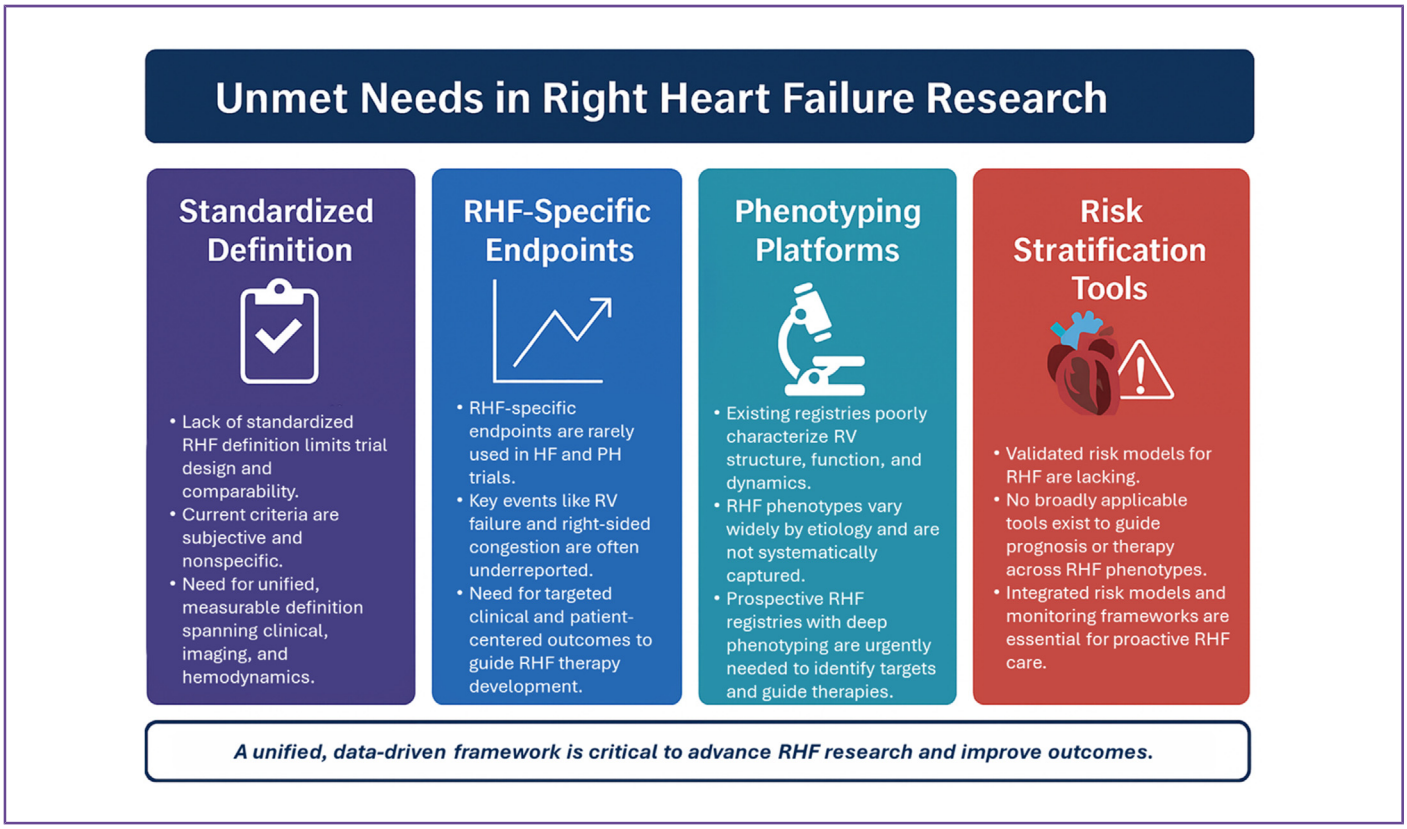


Fig 5. Unmet needs in research in the field of RHF. HF, heart failure; PH, pulmonary hypertension; RHF, right heart failure; RV, right ventricle.

underscoring the need for a measurable, multidimensional definition that integrates clinical, imaging, biomarker, and hemodynamic parameters.[177] Second, RHF-specific endpoints remain rarely incorporated in HF or PH studies, resulting in inconsistent capture of right-sided events such as RV failure or systemic congestion. Standardized, patient-centered RHF outcomes would substantially enhance the interpretability of clinical trials and registries.[148,178] Third, existing databases and registries lack adequate characterization of RV structure, function, and dynamics, highlighting the urgent need for prospective, deeply phenotyped RHF cohorts to identify targets and enable mechanistic discovery.[179] Finally, no validated, broadly applicable risk models exist to guide clinical decision-making or trial stratification across RHF etiologies.[180] The development of integrated, data-driven models that combine biomarkers, imaging indices, and invasive hemodynamics is critical to enable proactive, precision-based RHF care. Collectively, these deficits underscore the urgent need for a unified, data-driven framework to advance RHF research and improve outcomes.

Moving forward, a multipronged approach is required to overcome these gaps. RHF must be explicitly prioritized as a funding category by governmental agencies (eg, the National Institutes of Health), foundations, and industry sponsors.[181] Collaborative, cross-specialty engagement, particularly among HF, PH, critical care, cardiac surgery, and congenital heart disease communities, is essential to create unified frameworks and enable RHF-specific research. Future clinical trials should be designed to include RHF-enriched populations, RHF-specific endpoints, and stratified analyses on the basis of RV phenotype. Adaptive and platform trial methodologies may offer efficient strategies to evaluate multiple interventions across RHF subtypes. RHF remains one of the most understudied and misunderstood domains in cardiovascular medicine, yet it represents an extraordinary opportunity for scientific discovery. As the field advances toward more personalized, mechanism-driven care, defining, studying, and treating RHF with the same scientific rigor historically applied to left-sided HF will be critical. The path forward demands clarity of definition, consistency of measurement, innovation in monitoring, and a sustained commitment to collaborative investigation.

Disclosures

Manreet Kanwar is a consultant and speaker for Abbott and Abiomed/Johnson & Johnson MedTech and a CEC Member for BiVACOR. Anthony Carnicelli has research support from Acorai, honoraria and research support from Johnson & Johnson MedTech, and consulting fees from Precision Cardiovascular. James Coons is a consultant for AstraZeneca. Mark Belkin is a consultant for Bayer. Thomas M. Cascino reports (1) extramural support from the National Institutes of Health; (2) advisory board for CareDx, Impulse Dynamics, Johnson & Johnson, and Merck; and (3) speaker's bureau for Abbott, Inc. Ezequiel Molina is a consultant and speaker for Abbott and a CEC member for XVIVO. Ryan Tedford general disclosures include consulting relationships with and receiving honorarium/consulting fees from Abbott, Acorai, Adona, Aria CV Inc, Boston Scientific, CVRx, Endotronix, Edwards Lifesciences, Fauna Bio, Gradient, Imbria, Inhibikase, Merck, Morphic Therapeutics, Pulmovant, Restore Medical, Tempus AI, 35Pharma, and United Therapeutics. Dr Tedford serves on steering committees for Abbott, Edwards, Endotronix, Gradient, Merck, Restore Medical, and Tempus AI as well as a research advisory board for Abiomed.



Manreet K. Kanwar

CRediT authorship contribution statement

MANREET K. KANWAR: Conceptualization, Supervision, Writing – original draft, Writing – review & editing. **ANTHONY P. CARNICELLI:** Writing – original draft, Writing – review & editing. **MARK N. BELKIN:** Writing – original draft, Writing – review & editing. **MICHELE L. ESPOSITO:** Writing – original draft, Writing – review & editing. **JAMES C. COONS:** Writing – original draft, Writing – review & editing. **EZEQUIEL J. MOLINA:** Writing – original draft, Writing – review & editing. **SOPHIA AIRHART:** Writing – original draft, Writing – review & editing. **THOMAS M. CASCINO:** Writing – original draft, Writing – review & editing. **VANESSA BLUMER:** Writing – original draft, Writing – review & editing. **RYAN J. TEDFORD:** Writing – original draft, Writing – review & editing.

Funding

None.

Patient consent

The authors retrieved and synthesized data from previously published studies; therefore, no ethical approval was required or obtained.

References

1. Konstam MA, Kiernan MS, Bernstein D, Bozkurt B, Jacob M, Kapur NK, et al. Evaluation and management of right-sided heart failure: a scientific statement from the American Heart Association. Circulation 2018;137:e578–622.
2. Monteagudo-Vela M, Tindale A, Monguio-Santin E, Reyes-Copa G, Panoulas V. Right ventricular failure: current strategies and future development. Front Cardiovasc Med 2023;10:998382.
3. Lala A, Guo Y, Xu J, Esposito M, Morine K, Karas R, et al. Right ventricular dysfunction in acute myocardial infarction complicated by cardiogenic shock: a hemodynamic analysis of the Should We Emergently Revascularize Occluded Coronaries for Cardiogenic Shock (SHOCK) trial and registry. J Card Fail 2018;24:148–56.
4. Thayer KL, Zweck E, Ayouty M, Garan AR, Hernandez-Montfort J, Mahr C, et al. Invasive hemodynamic assessment and classification of in-hospital mortality risk among patients with cardiogenic shock. Circ Heart Fail 2020;13:e007099.
5. Tang WHW, Bakitas MA, Cheng XS, Fang JC, Fedson SE, Fiedler AG, et al. Evaluation and Management of kidney dysfunction in advanced heart failure: a scientific statement from the American Heart Association. Circulation 2024;150:e280–95.
6. Rosenkranz S, Howard LS, Gomberg-Maitland M, Hoeper MM. Systemic consequences of pulmonary hypertension and right-sided heart failure. Circulation 2020;141:678–93.
7. Anand IS, Gupta P. Anemia and iron deficiency in heart failure. Circulation 2018;138:80–98.
8. Benza RL, Kanwar MK, Raina A, Scott JV, Zhao CL, Selej M, et al. Development and validation of an abridged version of the REVEAL 2.0 risk score calculator, REVEAL Lite 2, for use in patients with pulmonary arterial hypertension. Chest 2021;159:337–46.
9. Barst RJ, Chung L, Zamanian RT, Turner M, McGoon MD. Functional class improvement and 3-year survival outcomes in patients with pulmonary arterial hypertension in the REVEAL registry. Chest 2013;144:160–8.
10. Ganipineni VDP, Jitta SR, Gudiwada M, Jasti JR, Janga C, Merugu B, et al. High-sensitivity cardiac troponin [hs-cTn] as a valuable biomarker for pulmonary hypertension risk stratification: a contemporary review of the literature. Healthcare (Basel) 2024;12:2037.
11. Heresi GA, Tang WH, Aytekin M, Hammel J, Hazen SL, Dweik RA. Sensitive cardiac troponin I predicts poor outcomes in pulmonary arterial hypertension. Eur Respir J 2012;39:939–44.
12. Snipelisky D, Jentzer J, Batal O, Dardari Z, Mathier M. Serum albumin concentration as an independent prognostic indicator in patients with pulmonary arterial hypertension. Clin Cardiol 2018;41:782–7.
13. Takeda Y, Takeda Y, Tomimoto S, Tani T, Narita H, Kimura G. Bilirubin as a prognostic marker in patients with pulmonary arterial hypertension. BMC Pulm Med 2010;10:22.
14. Shah SJ, Thenappan T, Rich S, Tian L, Archer SL, Gomberg-Maitland M. Association of serum creatinine with abnormal hemodynamics and mortality in pulmonary arterial hypertension. Circulation 2008;117:2475–83.
15. Yan L, Huang Z, Zhao Z, Zhao Q, Tang Y, Zhang Y, et al. The prognostic impact of serum uric acid on disease severity and 5-year mortality in patients with idiopathic pulmonary artery hypertension. Front Med (Lausanne) 2022;9:805415.
16. Uk Kang T, Park KY, Kim HJ, Ahn HS, Yim SY, Jun JB. Association of hyperuricemia and pulmonary hypertension: a systematic review and meta-analysis. Mod Rheumatol 2019;29:1031–41.
17. Forfia PR, Mathai SC, Fisher MR, Housten-Harris T, Hemnes AR, Champion HC, et al. Hyponatremia predicts right heart failure and poor survival in pulmonary arterial hypertension. Am J Respir Crit Care Med 2008;177:1364–9.
18. Bendayan D, Shitrit D, Ygla M, Huerta M, Fink G, Kramer MR. Hyperuricemia as a prognostic factor in pulmonary arterial hypertension. Respir Med 2003;97:130–3.
19. Hoeper MM, Pletz MW, Golpon H, Welte T. Prognostic value of blood gas analyses in patients with idiopathic pulmonary arterial hypertension. Eur Respir J 2007;29:944–50.
20. Mukherjee M, Rudski LG, Addetia K, Afilalo J, D'Alto M, Freed BH, et al. Guidelines for the echocardiographic assessment of the right heart in adults and special considerations in pulmonary hypertension: recommendations from the American Society of Echocardiography. J Am Soc Echocardiogr 2025;38:141–86.
21. El-Kersh K, Zhao C, Elliott G, Farber HW, Gomberg-Maitland M, Selej M, et al. Derivation of a risk score (REVEAL-ECHO) based on echocardiographic parameters of patients with pulmonary arterial hypertension. Chest 2023;163:1232–44.
22. Badagliacca R, Tello K, D'Alto M, Ghio S, Argiento P, Brunetti ND, et al. TAPSE/sPAP ratio to improve risk assessment in pulmonary arterial hypertension. Circ Heart Fail 2025;18:e012518.
23. Rommel KP, Schlotter F, Stolz L, Kresoja KP, Kassar M, Praz F, et al. Right ventricular-pulmonary artery coupling in tricuspid regurgitation: prognostic value and impact of treatment strategy. JACC Cardiovasc Interv 2025;18:1411–21.
24. Palacios-Moguel P, Cueto-Robledo G, Gonzalez-Pacheco H, Ortega-Hernandez J, Torres-Rojas MB, Navarro-Vergara DI, et al. The role of the TAPSE/sPAP ratio as a predictor of mortality in pulmonary arterial hypertension: its value for patient risk stratification. JHLT Open 2025;7:100168.
25. John T, Avian A, John N, Eger A, Foris V, Zeder K, et al. Prognostic relevance of tricuspid annular plane systolic excursion to systolic pulmonary arterial pressure ratio and its association with exercise hemodynamics in patients with normal or mildly elevated resting pulmonary arterial pressure. Chest 2025;167:573584.
26. Dardi F, Guarino D, Ballerini A, Bertozzi R, Donato F, Cennerazzo F, et al. Prognostic role of haemodynamics at follow-up in patients with pulmonary arterial hypertension: a challenge to current European Society of Cardiology/European Respiratory Society risk tools. ERJ Open Res. 2024;10:4.
27. Meucci MC, Malara S, Butcher SC, Hirasawa K, van der Kley F, Lombardo A, et al. Evolution and prognostic impact of right ventricular-pulmonary artery coupling after transcatheter aortic valve replacement. JACC Cardiovasc Interv 2023;16:1612–21.
28. Hahn RT, Lerakis S, Delgado V, Addetia K, Burkhoff D, Muraru D, et al. Multimodality imaging of right heart function: JACC Scientific Statement. J Am Coll Cardiol 2023;81:1954–73.
29. Colalillo A, Hoffmann-Vold AM, Pellicano C, Romaniello A, Gabrielli A, Hachulla E, et al. The role of TAPSE/sPAP ratio in predicting pulmonary hypertension and mortality in the systemic sclerosis EUSTAR cohort. Autoimmun Rev 2023;22:103290.
30. Humbert M, Kovacs G, Hoeper MM, Badagliacca R, Berger RMF, Brida M, et al. 2022 ESC/ERS guidelines for the diagnosis and treatment of pulmonary hypertension. Eur Heart J 2022;43:3618–731.
31. Ghio S, Mercurio V, Fortuni F, Forfia PR, Gall H, Ghofrani A, et al. A comprehensive echocardiographic method for risk stratification in pulmonary arterial hypertension. Eur Respir J 2020;56.
32. Tello K, Axmann J, Ghofrani HA, Naeije R, Narcin N, Rieth A, et al. Relevance of the TAPSE/PASP ratio in pulmonary arterial hypertension. Int J Cardiol 2018;266:229–35.
33. Bartolome SD, Sood N, Shah TG, Styrvoky K, Torres F, Chin KM. Mortality in patients with pulmonary arterial hypertension treated with continuous prostanoids. Chest 2018;154:532–40.
34. Badagliacca R, Poscia S, Pezzuto B, Papa S, Reali M, Pesce F, et al. Prognostic relevance of right heart reverse remodeling in idiopathic pulmonary arterial hypertension. J Heart Lung Transplant 2017;2:S1053-2498(17)32041−7.
35. Chen L, Larsen CM, Le RJ, Connolly HM, Pislaru SV, Murphy JG, et al. The prognostic significance of tricuspid valve regurgitation in pulmonary arterial hypertension. Clin Respir J 2018;12:1572–80.
36. Bosch L, Lam CSP, Gong L, Chan SP, Sim D, Yeo D, et al. Right ventricular dysfunction in left-sided heart failure with preserved versus reduced ejection fraction. Eur J Heart Fail 2017;19:1664–71.
37. Amsallem M, Sweatt AJ, Aymami MC, Kuznetsova T, Selej M, Lu H, et al. Right heart end-systolic remodeling index strongly predicts outcomes in pulmonary arterial hypertension: comparison with validated models. Circ Cardiovasc Imaging 2017;10:e005771.
38. Chin KM, Channick RN, Kim NH, Rubin LJ. Central venous blood oxygen saturation monitoring in patients with chronic pulmonary arterial hypertension treated with continuous IV epoprostenol: correlation with measurements of hemodynamics and plasma brain natriuretic peptide levels. Chest 2007;132:786–92.
39. Forfia PR, Fisher MR, Mathai SC, Housten-Harris T, Hemnes AR, Borlaug BA, et al. Tricuspid annular displacement predicts survival in pulmonary hypertension. Am J Respir Crit Care Med 2006;174:1034–41.
40. Raymond RJ, Hinderliter AL, Willis PW, Ralph D, Caldwell EJ, Williams W, et al. Echocardiographic predictors of adverse outcomes in primary pulmonary hypertension. J Am Coll Cardiol 2002;39:1214–9.
41. van der Bruggen CE, Handoko ML, Bogaard HJ, Marcus JT, Oosterveer FPT, Meijboom LJ, et al. The value of hemodynamic measurements or cardiac MRI in the follow-up of patients with idiopathic pulmonary arterial hypertension. Chest 2021;159:1575–85.
42. Brewis MJ, Bellofiore A, Vanderpool RR, Chesler NC, Johnson MK, Naeije R, et al. Imaging right ventricular function to predict outcome in pulmonary arterial hypertension. Int J Cardiol 2016;218:206–11.
43. Courand PY, Pina Jomir G, Khouatra C, Scheiber C, Turquier S, Glerant JC, et al. Prognostic value of right ventricular ejection fraction in pulmonary arterial hypertension. Eur Respir J 2015;45:139–49.
44. Gulati A, Ismail TF, Jabbour A, Alpendurada F, Guha K, Ismail NA, et al. The prevalence and prognostic significance of right ventricular systolic dysfunction in nonischemic dilated cardiomyopathy. Circulation 2013;128:1623–33.
45. van Wolferen SA, Marcus JT, Boonstra A, Marques KM, Bronzwaer JG, Spreeuwenberg MD, et al. Prognostic value of right ventricular mass, volume, and function in idiopathic pulmonary arterial hypertension. Eur Heart J 2007;28:1250–7.
46. Lewis RA, Johns CS, Cogliano M, Capener D, Tubman E, Elliot CA, et al. Identification of cardiac magnetic resonance imaging thresholds for risk stratification in pulmonary arterial hypertension. Am J Respir Crit Care Med 2020;201:458–68.
47. Keskin B, Hakgor A, Yilmaz B, Erkanli K, Cakal B, Yazar A, et al. Right atrial pressure/pulmonary capillary wedge pressure ratio predicts in-hospital mortality in left ventricular assist device recipients. J Clin Med 2025;14:4784.
48. Ozenc E, Yildiz O, Baydar O, Yazicioglu N, Koc NA. Impact of right ventricular stroke work index on predicting hospital readmission and functional status of patients with advanced heart failure. Rev Port Cardiol (Engl Ed) 2020;39:565–72.
49. Ibe T, Wada H, Sakakura K, Ito M, Ugata Y, Yamamoto K, et al. Right ventricular stroke work index. Int Heart J 2018;59:1047–51.
50. Kochav SM, Flores RJ, Truby LK, Topkara VK. Prognostic impact of pulmonary artery pulsatility index (PAPi) in patients with advanced heart failure: insights from the ESCAPE trial. J Card Fail 2018;24:453–9.
51. Csosza G, Valko L, Dinya E, Losonczy G, Muller V, Lazar Z, et al. Right ventricular stroke work index in pulmonary arterial hypertension and chronic thromboembolic pulmonary hypertension: a retrospective observational study. Pulm Circ 2024;14:e12433.
52. Armstrong HF, Schulze PC, Kato TS, Bacchetta M, Thirapatarapong W, Bartels MN. Right ventricular stroke work index as a negative predictor of mortality and initial hospital stay after lung transplantation. J Heart Lung Transplant 2013;32:603–8.

53. Fitzpatrick 3rd JR, Frederick JR, Hsu VM, Kozin ED, O'Hara ML, Howell E, et al. Risk score derived from pre-operative data analysis predicts the need for biventricular mechanical circulatory support. J Heart Lung Transplant 2008;27:1286–92.
54. Suzuki S, Asano R, Aoki T, Nakayama S, Ueda J, Tsuji A, et al. Prognostic impact of follow-up pulmonary vascular resistance in pulmonary arterial hypertension. Open Heart 2022;9:e002054.
55. Benza RL, Miller DP, Gomberg-Maitland M, Frantz RP, Foreman AJ, Coffey CS, et al. Predicting survival in pulmonary arterial hypertension: insights from the Registry to Evaluate Early and Long-Term Pulmonary Arterial Hypertension Disease Management (REVEAL). Circulation 2010;122:164–72.
56. Fares WH, Bellumkonda L, Tonelli AR, Carson SS, Hassoun PM, Trow TK, et al. Right atrial pressure/pulmonary artery wedge pressure ratio: a more specific predictor of survival in pulmonary arterial hypertension. J Heart Lung Transplant 2016;35:760–7.
57. Rayner SG, Tedford RJ, Leary PJ, Mak S, Houston BA. "This patient needs a doctor, not a guideline!" The zone of uncertainty in pulmonary arterial wedge pressure measurement. Am J Respir Crit Care Med 2024;210:712–4.
58. Reddy YNV, Frantz RP, Miranda WR, Asokan AK, Varma R, Rischard F, et al. Elevated pulmonary artery wedge pressure in group 1 pulmonary hypertension. Circulation 2026;153:948–62.
59. Viray MC, Bonno EL, Gabrielle ND, Maron BA, Atkins J, Amoroso NS, et al. Role of pulmonary artery wedge pressure saturation during right heart catheterization: a prospective Study. Circ Heart Fail 2020;13:e007981.
60. Yim IHW, Khan-Kheil AM, Drury NE, Lim HS. A systematic review and physiology of pulmonary artery pulsatility index in left ventricular assist device therapy. Interdiscip Cardiovasc Thorac Surg 2023;36:ivad068.
61. Zern EK, Wang D, Rambarat P, Bernard S, Paniagua SM, Liu EE, et al. Association of pulmonary artery pulsatility index with adverse cardiovascular events across a hospital-based sample. Circ Heart Fail 2022;15:e009085.
62. Lim HS, Gustafsson F. Pulmonary artery pulsatility index: physiological basis and clinical application. Eur J Heart Fail 2020;22:32–8.
63. Kang G, Ha R, Banerjee D. Pulmonary artery pulsatility index predicts right ventricular failure after left ventricular assist device implantation. J Heart Lung Transplant 2016;35:67–73.
64. Korabathina R, Heffernan KS, Paruchuri V, Patel AR, Mudd JO, Prutkin JM, et al. The pulmonary artery pulsatility index identifies severe right ventricular dysfunction in acute inferior myocardial infarction. Catheter Cardiovasc Interv 2012;80:593–600.
65. Morine KJ, Kiernan MS, Pham DT, Paruchuri V, Denofrio D, Kapur NK. Pulmonary artery pulsatility index is associated with right ventricular failure after left ventricular assist device surgery. J Card Fail 2016;22:110–6.
66. Mazimba S, Welch TS, Mwansa H, Breathett KK, Kennedy JLW, Mihalek AD, et al. Haemodynamically derived pulmonary artery pulsatility index predicts mortality in pulmonary arterial hypertension. Heart Lung Circ 2019;28:752–60.
67. Cesini S, Bhagra S, Pettit SJ. Low pulmonary artery pulsatility index is associated with adverse outcomes in ambulatory patients with advanced heart failure. J Card Fail 2020;26:352–9.
68. Richter MJ, Hsu S, Yogeswaran A, Husain-Syed F, Vadasz I, Ghofrani HA, et al. Right ventricular pressure-volume loop shape and systolic pressure change in pulmonary hypertension. Am J Physiol Lung Cell Mol Physiol 2021;320:L715–25.
69. Sahay S, Visovatti S, Tonelli AR, Villasmil Hernandez N, Austin ED, Badagliacca R, et al. International Society for Heart and Lung Transplantation (ISHLT) consensus statement on risk stratification in pulmonary arterial hypertension. J Heart Lung Transplant 2025;44:e73–e131.
70. Benza RL, Gomberg-Maitland M, Elliott CG, Farber HW, Foreman AJ, Frost AE, et al. Predicting survival in patients with pulmonary arterial hypertension: the REVEAL Risk Score Calculator 2.0 and comparison with ESC/ERS-based risk assessment strategies. Chest 2019;156:323–7.
71. Benza RL, Miller DP, Barst RJ, Badesch DB, Frost AE, McGoon MD. An evaluation of long-term survival from time of diagnosis in pulmonary arterial hypertension from the REVEAL Registry. Chest 2012;142:448–56.
72. Taleb I, Kyriakopoulos CP, Fong R, Ijaz N, Demertzis Z, Sideris K, et al. Machine learning multicenter risk model to predict right ventricular failure after mechanical circulatory support: the STOP-RVF Score. JAMA Cardiol 2024;9:272–82.
73. Matthews JC, Koelling TM, Pagani FD, Aaronson KD. The right ventricular failure risk score a pre-operative tool for assessing the risk of right ventricular failure in left ventricular assist device candidates. J Am Coll Cardiol 2008;51:2163–72.
74. Soliman OII, Akin S, Muslem R, Boersma E, Manintveld OC, Krabatsch T, et al. Derivation and validation of a novel right-sided heart failure model after implantation of continuous flow left ventricular assist devices: the EUROMACS (European Registry for Patients with Mechanical Circulatory Support) right-sided heart failure risk score. Circulation 2018;137:891–906.
75. Tchantchaleishvili V, Maltais S, Sharma S, Haglund NA, Davis ME, Cowger J, et al. A novel, highly discriminatory risk model predicting acute severe right ventricular failure in patients undergoing continuous-flow left ventricular assist device implant. Artif Organs 2019;43:624–32.
76. Loforte A, Montalto A, Musumeci F, Amarelli C, Mariani C, Polizzi V, et al. Calculation of the ALMA risk of right ventricular failure after left ventricular assist device implantation. ASAIO J 2018;64:e140–7.
77. Atluri P, Goldstone AB, Fairman AS, MacArthur JW, Shudo Y, Cohen JE, et al. Predicting right ventricular failure in the modern, continuous flow left ventricular assist device era. Ann Thorac Surg 2013;96:857–63. discussion 863−4.
78. Drakos SG, Janicki L, Horne BD, Kfoury AG, Reid BB, Clayson S, et al. Risk factors predictive of right ventricular failure after left ventricular assist device implantation. Am J Cardiol 2010;105:1030–5.
79. Kormos RL, Teuteberg JJ, Pagani FD, Russell SD, John R, Miller LW, et al. Right ventricular failure in patients with the HeartMate II continuous-flow left ventricular assist device: incidence, risk factors, and effect on outcomes. J Thorac Cardiovasc Surg 2010;139:1316–24.
80. Tsuji M, Kurihara T, Isotani Y, Bujo C, Ishida J, Amiya E, et al. Right heart reserve function assessed with fluid loading predicts late right heart failure after left ventricular assist device implantation. Can J Cardiol 2024;40:1745–54.
81. Baratto C, Caravita S, Dewachter C, Faini A, Perego GB, Bondue A, et al. Right heart adaptation to exercise in pulmonary hypertension: an invasive hemodynamic study. J Card Fail 2023;29:1261–72.
82. Caravita S, Baratto C, Filippo A, Soranna D, Dewachter C, Zambon A, et al. Shedding light on latent pulmonary vascular disease in heart failure with preserved ejection fraction. JACC Heart Fail 2023;11:1427–38.
83. Al-Qadi MO, Holbrook J, Ford HJ, Ceppe A, LeVarge BL. Prognostic value of respiratory variation in right atrial pressure in patients with precapillary pulmonary hypertension. Chest 2023;164:481–9.
84. Read JM, Azih NI, Peters CJ, Gurtu V, Vishram-Nielsen JK, Wright SP, et al. Hemodynamic reserve predicts early right heart failure after LVAD implantation. J Heart Lung Transplant 2022;41:1716–26.
85. Nagel C, Marra AM, Benjamin N, Blank N, Cittadini A, Coghlan G, et al. Reduced right ventricular output reserve in patients with systemic sclerosis and mildly elevated pulmonary artery pressure. Arthritis Rheumatol 2019;71:805–16.
86. Brinkley Jr. DM, Ho KKL, Drazner MH, Kociol RD. The prognostic value of the relationship between right atrial and pulmonary capillary wedge pressure in diverse cardiovascular conditions. Am Heart J 2018;199:31–6.
87. Chung YW, Kim HK, Kim IY, Yim MB, Chock PB. Dual function of protein kinase C (PKC) in 12-O-tetradecanoylphorbol-13-acetate (TPA)-induced manganese superoxide dismutase (MnSOD) expression: activation of CREB and FOXO3a by PKC-alpha phosphorylation and by PKC-mediated inactivation of Akt, respectively. J Biol Chem 2011;286:29681–90.
88. Drazner MH, Hamilton MA, Fonarow G, Creaser J, Flavell C, Stevenson LW. Relationship between right and left-sided filling pressures in 1000 patients with advanced heart failure. J Heart Lung Transplant 1999;18:1126–32.
89. Labrada L, Romero C, Sadek A, Belardo D, Raza Y, Forfia P. Intravenous diuresis in severe precapillary pulmonary-hypertension-related right heart failure: effects on renal function and blood pressure. J Clin Med 2023;12:7149.
90. Stickel S, Gin-Sing W, Wagenaar M, Gibbs JSR. The practical management of fluid retention in adults with right heart failure due to pulmonary arterial hypertension. Eur Heart J Suppl 2019;21:K46–53.
91. Savale L, Benazzo A, Corris P, Keshavjee S, Levine DJ, Mercier O, et al. Transplantation, bridging, and support technologies in pulmonary hypertension. Eur Respir J 2024;64:2401193.
92. Hansen L, Burks M, Kingman M, Stewart T. Volume management in pulmonary arterial hypertension patients: an expert pulmonary hypertension clinician perspective. Pulm Ther 2018;4:13–27.
93. Wilcox CS, Testani JM, Pitt B. Pathophysiology of diuretic resistance and its implications for the management of chronic heart failure. Hypertension 2020;76:1045–54.
94. Valente MA, Voors AA, Damman K, Van Veldhuisen DJ, Massie BM, O'Connor CM, et al. Diuretic response in acute heart failure: clinical characteristics and prognostic significance. Eur Heart J 2014;35:1284–93.
95. Felker GM, Ellison DH, Mullens W, Cox ZL, Testani JM. Diuretic therapy for patients with heart failure: JACC State-of-the-Art Review. J Am Coll Cardiol 2020;75:1178–95.
96. Marenzi G, Muratori M, Cosentino ER, Rinaldi ER, Donghi V, Milazzo V, et al. Continuous ultrafiltration for congestive heart failure: the CUORE trial. J Card Fail 2014;20:9–17.
97. Bart BA, Goldsmith SR, Lee KL, Givertz MM, O'Connor CM, Bull DA, et al. Ultrafiltration in decompensated heart failure with cardiorenal syndrome. N Engl J Med 2012;367:2296–304.
98. Giglioli C, Landi D, Cecchi E, Chiostri M, Gensini GF, Valente S, et al. Effects of ULTRAfiltration vs. DIureticS on clinical, biohumoral and haemodynamic variables in patients with deCOmpensated heart failure: the ULTRADISCO study. Eur J Heart Fail 2011;13:337–46.
99. Costanzo MR, Guglin ME, Saltzberg MT, Jessup ML, Bart BA, Teerlink JR, et al. Ultrafiltration versus intravenous diuretics for patients hospitalized for acute decompensated heart failure. J Am Coll Cardiol 2007;49:675–83.
100. Thangapazham RL, Singh AK, Sharma A, Warren J, Gaddipati JP, Maheshwari RK. Green tea polyphenols and its constituent epigallocatechin gallate inhibits proliferation of human breast cancer cells in vitro and in vivo. Cancer Lett 2007;245:232–41.
101. Costanzo MR, Saltzberg M, O'Sullivan J, Sobotka P. Early ultrafiltration in patients with decompensated heart failure and diuretic resistance. J Am Coll Cardiol 2005;46:2047–51.
102. Bart BA, Boyle A, Bank AJ, Anand I, Olivari MT, Kraemer M, et al. Ultrafiltration versus usual care for hospitalized patients with heart failure: the Relief for Acutely Fluid-Overloaded Patients With Decompensated Congestive Heart Failure (RAPID-CHF) trial. J Am Coll Cardiol 2005;46:2043–6.
103. Sabato LA, Salerno DM, Moretz JD, Jennings DL. Inhaled pulmonary vasodilator therapy for management of right ventricular dysfunction after left ventricular assist device placement and cardiac transplantation. Pharmacotherapy 2017;37:944–55.
104. Sitbon O, Jaïs X, Savale L, Cottin V, Bergot E, Macari EA, et al. Upfront triple combination therapy in pulmonary arterial hypertension: a pilot study. Eur Respir J 2014;43:1691–7.
105. Fischer LG, Aken HV, Bürkle H. Management of pulmonary hypertension: physiological and pharmacological considerations for anesthesiologists. Anesth Analg 2003;96:1603–16.
106. Solina AR, Ginsberg SH, Papp D, Grubb WR, Scholz PM, Pantin EJ, et al. Dose response to nitric oxide in adult cardiac surgery patients. J Clin Anesth 2001;13:281–6.

107. Argenziano M, Choudhri AF, Moazami N, Rose EA, Smith CR, Levin HR, et al. Randomized, double-blind trial of inhaled nitric oxide in LVAD recipients with pulmonary hypertension. Ann Thorac Surg 1998;65:340–5.
108. Hoeper MM, Benza RL, Corris P, de Perrot M, Fadel E, Keogh AM, et al. Intensive care, right ventricular support and lung transplantation in patients with pulmonary hypertension. Eur Respir J 2019;53:1801906.
109. Haraldsson A, Kieler-Jensen N, Wadenvik H, Ricksten SE. Inhaled prostacyclin and platelet function after cardiac surgery and cardiopulmonary bypass. Intensive Care Med 2000;26:188–94.
110. Fowler MB, Alderman EL, Oesterle SN, Derby G, Daughters GT, Stinson EB, et al. Dobutamine and dopamine after cardiac surgery: greater augmentation of myocardial blood flow with dobutamine. Circulation 1984;70:I103–11.
111. Jentzer JC, Coons JC, Link CB, Schmidhofer M. Pharmacotherapy update on the use of vasopressors and inotropes in the intensive care unit. J Cardiovasc Pharmacol Ther 2015;20:249–60.
112. Aranda Jr. JM, Schofield RS, Pauly DF, Cleeton TS, Walker TC, Monroe Jr. VS, et al. Comparison of dobutamine versus milrinone therapy in hospitalized patients awaiting cardiac transplantation: a prospective, randomized trial. Am Heart J 2003;145:324–9.
113. Chen EP, Bittner HB, Davis Jr. RD, Van Trigt 3rd P. Milrinone improves pulmonary hemodynamics and right ventricular function in chronic pulmonary hypertension. Ann Thorac Surg 1997;63:814–21.
114. Metra M, Nodari S, D'Aloia A, Muneretto C, Robertson AD, Bristow MR, et al. Beta-blocker therapy influences the hemodynamic response to inotropic agents in patients with heart failure: a randomized comparison of dobutamine and enoximone before and after chronic treatment with metoprolol or carvedilol. J Am Coll Cardiol 2002;40:1248–58.
115. Russell JA, Gordon AC, Williams MD, Boyd JH, Walley KR, Kissoon N. Vasopressor therapy in the intensive care unit. Semin Respir Crit Care Med 2021;42:59–77.
116. Rich S, Seidlitz M, Dodin E, Osimani D, Judd D, Genthner D, et al. The short-term effects of digoxin in patients with right ventricular dysfunction from pulmonary hypertension. Chest 1998;114:787–92.
117. Davis JO, Howell DS, Hyatt RE. Effects of acute and chronic digoxin administration in dogs with right-sided congestive heart failure produced by pulmonary artery constriction. Circ Res 1955;3:259–63.
118. Chang KY, Giorgio K, Schmitz K, Walker RF, Prins KW, Pritzker MR, et al. Effect of chronic digoxin use on mortality and heart failure hospitalization in pulmonary arterial hypertension. J Am Heart Assoc 2023;12:e027559.
119. Andersen S, Andersen A, de Man FS, Nielsen-Kudsk JE. Sympathetic nervous system activation and beta-adrenoceptor blockade in right heart failure. Eur J Heart Fail 2015;17:358–66.
120. Soulat-Dufour L, Lang S, Addetia K, Ederhy S, Adavane-Scheuble S, Chauvet-Droit M, et al. Restoring sinus rhythm reverses cardiac remodeling and reduces valvular regurgitation in patients with atrial fibrillation. J Am Coll Cardiol 2022;79:951–61.
121. Arrigo M, Jaeger N, Seifert B, Spahn DR, Bettex D, Rudiger A. Disappointing success of electrical cardioversion for new-onset atrial fibrillation in cardiosurgical ICU patients. Crit Care Med 2015;43:2354–9.
122. Kholdani CA, Fares WH. Management of right heart failure in the intensive care unit. Clin Chest Med 2015;36:511–20.
123. King C, May CW, Williams J, Shlobin OA. Management of right heart failure in the critically ill. Crit Care Clin 2014;30:475–98.
124. Harvey RM, Enson Y, Betti R, Lewis ML, Rochester DF, Ferrer MI. Further observations on the effect of hydrogen ion on the pulmonary circulation. Circulation 1967;35:1019–27.
125. Rudolph AM, Yuan S. Response of the pulmonary vasculature to hypoxia and H+ ion concentration changes. J Clin Invest 1966;45:399–411.
126. Yuriditsky E, Mireles-Cabodevila E, Alviar CL. How I teach: heart–lung interactions during mechanical ventilation. Positive pressure and the right ventricle. ATS Scholar 2024;6:94–108.
127. Alviar CL, Miller PE, McAreavey D, Katz JN, Lee B, Moriyama B, et al. Positive pressure ventilation in the cardiac intensive care unit. J Am Coll Cardiol 2018;72:1532–53.
128. Cheifetz IM, Craig DM, Quick G, McGovern JJ, Cannon ML, Ungerleider RM, et al. Increasing tidal volumes and pulmonary overdistention adversely affect pulmonary vascular mechanics and cardiac output in a pediatric swine model. Crit Care Med 1998;26:710–6.
129. Howell JB, Permutt S, Proctor DF, Riley RL. Effect of inflation of the lung on different parts of pulmonary vascular bed. J Appl Physiol 1961;16:71–6.
130. Masip J, Peacock WF, Price S, Cullen L, Martin-Sanchez FJ, Seferovic P, et al. Indications and practical approach to non-invasive ventilation in acute heart failure. Eur Heart J 2018;39:17–25.
131. Casey JD, Seitz KP, Driver BE, Gibbs KW, Ginde AA, Trent SA, et al. Ketamine or etomidate for tracheal intubation of critically ill adults. N Engl J Med 2026;394:1608–20.
132. Ahmad I, El-Boghdadly K, Bhagrath R, Hodzovic I, McNarry AF, Mir F, et al. Difficult Airway Society guidelines for awake tracheal intubation (ATI) in adults. Anaesthesia 2020;75:509–28.
133. DeFilippis EM, Topkara VK, Kirtane AJ, Takeda K, Naka Y, Garan AR. Mechanical circulatory support for right ventricular failure. Card Fail Rev 2022;8:e14.
134. Carnicelli AP, Diepen SV, Gage A, Bernhardt AM, Cowger J, Houston BA, et al. Pragmatic approach to temporary mechanical circulatory support in acute right ventricular failure. J Heart Lung Transplant 2024;43:1894–904.
135. Peled Y, Ducharme A, Kittleson M, Bansal N, Stehlik J, Amdani S, et al. International Society for Heart and Lung Transplantation guidelines for the evaluation and care of cardiac transplant candidates—2024. J Heart Lung Transplant 2024;43. 1529–28. e54.
136. Rashkind WJ, Miller WW. Creation of an atrial septal defect without thoracotomy. A palliative approach to complete transposition of the great arteries. JAMA 1966;196:991–2.
137. Rich S, Lam W. Atrial septostomy as palliative therapy for refractory primary pulmonary hypertension. Am J Cardiol 1983;51:1560–1.
138. Sandoval J, Rothman A, Pulido T. Atrial septostomy for pulmonary hypertension. Clin Chest Med 2001;22:547–60.
139. McIntyre KM, Sasahara AA. The hemodynamic response to pulmonary embolism in patients without prior cardiopulmonary disease. Am J Cardiol 1971;28:288–94.
140. Zuin M, Bikdeli B, Ballard-Hernandez J, Barco S, Battinelli EM, Giannakoulas G, et al. International clinical practice guideline recommendations for acute pulmonary embolism: harmony, dissonance, and silence. J Am Coll Cardiol 2024;84:1561–77.
141. Konstantinides SV, Meyer G, Becattini C, Bueno H, Geersing GJ, Harjola VP, et al. 2019 ESC guidelines for the diagnosis and management of acute pulmonary embolism developed in collaboration with the European Respiratory Society (ERS). Eur Heart J 2020;41:543–603.
142. Leiva O, Alviar C, Khandhar S, Parikh SA, Toma C, Postelnicu R, et al. Catheter-based therapy for high-risk or intermediate-risk pulmonary embolism: death and re-hospitalization. Eur Heart J 2024;45:1988–98.
143. Toma C, Bunte MC, Cho KH, Jaber WA, Chambers J, Stegman B, et al. Percutaneous mechanical thrombectomy in a real-world pulmonary embolism population: Interim results of the FLASH registry. Catheter Cardiovasc Interv 2022;99:1345–55.
144. Pei DT, Liu J, Yaqoob M, Ahmad W, Bandeali SS, Hamzeh IR, et al. Meta-analysis of catheter directed ultrasound-assisted thrombolysis in pulmonary embolism. Am J Cardiol 2019;124:1470–7.
145. Piazza G, Hohlfelder B, Jaff MR, Ouriel K, Engelhardt TC, Sterling KM, et al. A prospective, single-arm, multicenter trial of ultrasound-facilitated, catheter-directed, low-dose fibrinolysis for acute massive and submassive pulmonary embolism: the SEATTLE II study. JACC Cardiovasc Interv 2015;8:1382–92.
146. Kobashigawa J, Zuckermann A, Macdonald P, Leprince P, Esmailian F, Luu M, et al. Report from a consensus conference on primary graft dysfunction after cardiac transplantation. J Heart Lung Transplant 2014;33:327–40.
147. Velleca A, Shullo MA, Dhital K, Azeka E, Colvin M, DePasquale E, et al. The International Society for Heart and Lung Transplantation (ISHLT) guidelines for the care of heart transplant recipients. J Heart Lung Transplant 2023;42:e1–e141.
148. Grinstein J, Belkin MN, Kalantari S, Bourque K, Salerno C, Pinney S. Adverse hemodynamic consequences of continuous left ventricular mechanical support: JACC Review Topic of the Week. J Am Coll Cardiol 2023;82:70–81.
149. Kanelidis AJ, Gozdecki L, Belkin MN, Kalantari S, Nguyen A, Chung BB, et al. Mitigating post-operative right ventricular dysfunction after left ventricular assist device: the RV Protection Study. J Card Fail 2026;32:46–57.
150. Xanthopoulos A, Wolski K, Wang Q, Blackstone EH, Randhawa VK, Soltesz EG, et al. Postimplant phosphodiesterase-5 inhibitor use in centrifugal flow left ventricular assist devices. JACC Heart Fail 2022;10:89–100.
151. Monzo L, Reichenbach A, Al-Hiti H, Borlaug BA, Havlenova T, Solar N, et al. Acute unloading effects of sildenafil enhance right ventricular-pulmonary artery coupling in heart failure. J Card Fail 2021;27:224–32.
152. Tedford RJ, Hemnes AR, Russell SD, Wittstein IS, Mahmud M, Zaiman AL, et al. PDE5A inhibitor treatment of persistent pulmonary hypertension after mechanical circulatory support. Circ Heart Fail 2008;1:213–9.
153. Frantz RP, Desai SS, Ewald G, Franco V, Hage A, Horn EM, et al. SOPRANO: macitentan in patients with pulmonary hypertension following left ventricular assist device implantation. Pulm Circ 2024;14:e12446.
154. Estrada VH, Franco DL, Moreno AA, Gambasica JA, Nunez CC. Postoperative right ventricular failure in cardiac surgery. Cardiol Res 2016;7:185–95.
155. Chin KM, Gaine SP, Gerges C, Jing ZC, Mathai SC, Tamura Y, et al. Treatment algorithm for pulmonary arterial hypertension. Eur Respir J 2024;64:2401325.
156. Maron BA, Bortman G, De Marco T, Huston JH, Lang IM, Rosenkranz SH, et al. Pulmonary hypertension associated with left heart disease. Eur Respir J 2024;64:2401344.
157. Heidenreich PA, Bozkurt B, Aguilar D, Allen LA, Byun JJ, Colvin MM, et al. 2022 AHA/ACC/HFSA guideline for the management of heart failure: a report of the American College of Cardiology/American Heart Association Joint Committee on Clinical Practice Guidelines. Circulation 2022;145:e895–e1032.
158. Kittleson MM, Panjrath GS, Amancherla K, Davis LL, Deswal A, Dixon DL, et al. 2023 ACC expert consensus decision pathway on management of heart failure with preserved ejection fraction. JACC 2023;81:1835–78.
159. Borlaug BA, Reddy YNV, Braun A, Sorimachi H, Omar M, Popovic D, et al. Cardiac and metabolic effects of dapagliflozin in heart failure with preserved ejection fraction: the CAMEO-DAPA trial. Circulation 2023;148:834–44.
160. Cinar T, Saylik F, Cicek V, Pay L, Khachatryan A, Alejandro J, et al. Effects of SGLT2 inhibitors on right ventricular function in heart failure patients: updated meta-analysis of the current literature. Kardiol Pol 2024;82:416–22.
161. Nassif ME, Qintar M, Windsor SL, Jermyn R, Shavelle DM, Tang F, et al. Empagliflozin effects on pulmonary artery pressure in patients with heart failure. Circulation 2021;143:1673–86.
162. Gorter TM, van Veldhuisen DJ, Bauersachs J, Borlaug BA, Celutkiene J, Coats AJS, et al. Right heart dysfunction and failure in heart failure with preserved ejection fraction: mechanisms and management. Position statement on behalf of the Heart Failure Association of the European Society of Cardiology. Eur J Heart Fail 2018;20:16–37.

ARTICLE IN PRESS

163. Armstrong PW, Pieske B, Anstrom KJ, Ezekowitz J, Hernandez AF, Butler J, et al. Vericiguat in patients with heart failure and reduced ejection fraction. N Engl J Med 2020;382:1883–93.
164. Gomberg-Maitland M, Tedford RJ, Langleben D, Rosenkranz S, Miller B, Jones AD, et al. Sotatercept for Combined Post- and Pre-capillary Pulmonary Hypertension Associated With Heart Failure: Results from the Phase 2, Randomized, Placebo-Controlled CADENCE Study. Circulation 2026;153:19.
165. Kim NH, D'Armini AM, Delcroix M, Jais X, Jevnikar M, Madani MM, et al. Chronic thromboembolic pulmonary disease. Eur Respir J 2024;64:2401294.
166. de Perrot M, Donahoe L, McRae K, Thenganatt J, Moric J, Chan J, et al. Outcome after pulmonary endarterectomy for segmental chronic thromboembolic pulmonary hypertension. J Thorac Cardiovasc Surg 2022;164:696–707. e4.
167. Hsieh WC, Jansa P, Huang WC, Niznansky M, Omara M, Lindner J. Residual pulmonary hypertension after pulmonary endarterectomy: a meta-analysis. J Thorac Cardiovasc Surg 2018;156:1275–87.
168. Humbert M, Kovacs G, Hoeper MM, Badagliacca R, Berger RMF, Brida M, et al. 2022 ESC/ERS Guidelines for the diagnosis and treatment of pulmonary hypertension. Eur Respir J 2023;61:2200879.
169. Benza RL, Ghofrani HA, Grunig E, Hoeper MM, Jansa P, Jing ZC, et al. Effect of riociguat on right ventricular function in patients with pulmonary arterial hypertension and chronic thromboembolic pulmonary hypertension. J Heart Lung Transplant 2021;40:1172–80.
170. Gossl M, Stanberry L, Benson G, Steele E, Garberich R, Witt D, et al. Burden of undiagnosed valvular heart disease in the elderly in the community: Heart of New Ulm Valve Study. JACC Cardiovasc Imaging 2023;16:1118–20.
171. Hahn RT, Lawlor MK, Davidson CJ, Badhwar V, Sannino A, Spitzer E, et al. Tricuspid Valve Academic Research Consortium definitions for tricuspid regurgitation and trial endpoints. J Am Coll Cardiol 2023;82:1711–35.
172. Maisano F, Hahn R, Sorajja P, Praz F, Lurz P. Transcatheter treatment of the tricuspid valve: current status and perspectives. Eur Heart J 2024;45:876–94.
173. Zack CJ, Fender EA, Chandrashekar P, Reddy YNV, Bennett CE, Stulak JM, et al. National trends and outcomes in isolated tricuspid valve surgery. J Am Coll Cardiol 2017;70:2953–60.
174. Neuhold S, Huelsmann M, Pernicka E, Graf A, Bonderman D, Adlbrecht C, et al. Impact of tricuspid regurgitation on survival in patients with chronic heart failure: unexpected findings of a long-term observational study. Eur Heart J 2013;34:844–52.
175. Sorajja P, Whisenant B, Hamid N, Naik H, Makkar R, Tadros P, et al. Transcatheter repair for patients with tricuspid regurgitation. N Engl J Med 2023;388:1833–42.
176. Hahn RT, Makkar R, Thourani VH, Makar M, Sharma RP, Haeffele C, et al. Transcatheter valve replacement in severe tricuspid regurgitation. N Engl J Med 2025; 392:115–26.
177. Abraham WT, Psotka MA, Fiuzat M, Filippatos G, Lindenfeld J, Mehran R, et al. Standardized definitions for evaluation of heart failure therapies: scientific expert panel from the Heart Failure Collaboratory and Academic Research Consortium. Eur J Heart Fail 2020;22:2175–86.
178. Grinstein J, Cowger JA, Belkin MN, Houston BA, Tedford RJ. Hemodynamic consequences of long-term continuous flow: the importance of the right ventricular-aortic valve interactions. Circ Heart Fail 2023;16:e010713.
179. Jogendran R, Upagupta C, Pankiw M, Sless RT, Karvasarski E, Granton JT, et al. Toward a universal definition for right heart failure: a work in progress. Can J Cardiol 2025;41:1152–61.
180. Chaudhary RK, Ali O, Kumar A, Kumar A, Rab SM. Empagliflozin and right-sided heart failure: a comprehensive review of emerging evidence and clinical perspectives. Cureus 2025;17:e84064.
181. Leopold JA, Kawut SM, Aldred MA, Archer SL, Benza RL, Bristow MR, et al. Diagnosis and treatment of right heart failure in pulmonary vascular diseases: a National Heart, Lung, and Blood Institute workshop. Circ Heart Fail 2021;14:e007975.
182. Han J, Siddiki M, Kalantari S, Belkin MN, Grinstein J. Advanced hemodynamics for right ventricular failure: Evaluating the use of the myocardial performance score. J Heart Lung Transplant 2026;45(3):378–85.